The Resilience and Reinforcement of Group Cohesion in the Face of Economic Inequality

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Abstract: Building on evidence that group cohesion drives economic efficiency, we hypothesise income inequality influences efficiency by altering cohesion, with the direction depending on inequality type. Using a novel experimental design with extensive group interactions and a state-of-the-art group cohesion measure, we confirm stronger cohesion is associated with more efficient interactions. Group cohesion is strengthened by inequality *between* groups, a mechanism previously overlooked. Furthermore, unexpectedly, inequalities *within* groups do *not* hinder cohesion. These results suggest interpersonal relationships display resilience to economic disparities, challenging the prevailing argument that inequality must harm cohesion, and undermining a pathway through which it could reduce efficiency.

JEL Codes: C99, D31, D64, D91

Keywords: economic inequality; group cohesion; social identity; cooperation; coordination; public goods; weak-link game; economic experiment

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1. Introduction

The relationship between equality and efficiency is of paramount importance in economics. While traditional thinking points towards inequality being a necessary evil to facilitate efficiency, a growing school of thought contends that it in fact has the opposite effect of damaging efficiency. This may occur if a highly unequal society becomes a breeding ground for economically costly social problems. More unequal countries are found to have lower levels of trust, higher crime and corruption rates, and higher levels of conflict and instability (e.g. Alesina and Perotti, 1996; Alesina and La Ferrara, 2002; Wilkinson and Pickett, 2010; Olaoye et al., 2025). Relatedly, it has been argued that inequality may lead to a less cohesive society (e.g. Stiglitz, 2012; Xu and Marandola, 2023), characterized by a lower quality of social relationships between individuals who need to interact in the process of economic production.

Motivated by this reasoning, the current paper empirically investigates how inequality affects the cohesiveness of social relationships, and consequently economic efficiency. We build upon recent economic research identifying *group cohesion* as a meaningful factor in the productive process (Gächter et al., 2025). These findings support longstanding ideas in management, as well as in folk wisdom, that teams whose members enjoy closer interpersonal relationships work more efficiently together (Evans and Dion, 1991; Forsyth, 2021). If so, any factor - such as inequality - that influences group cohesion is likely to also affect economic efficiency.

Our empirical design was informed by a set of hypotheses, the testing of which ultimately led to a mixture of vindication and surprise. Prior to data collection, and expanding upon the arguments above, we conjectured that inequality *can* harm cohesion, but that this depends on the nature of the inequality. In the case of high inequality of earnings *within* groups, our expectation was that resentment would fester and the development of group cohesion would be relatively stunted. Conversely, when inequality comprises the difference in earnings *between* different groups, we hypothesised this would enhance the perceived distinctiveness of groups, breeding solidarity and therein contributing towards stronger group cohesion.

To test for *causal* effects on efficiency via the channel of group cohesion, we employed an experiment with several novel design features. We created groups of four people who interacted extensively and meaningfully over a half-day programme. Unlike in typical lab experiments, these interactions were face-to-face and included a combination of work and social activities similar to those found in real organisations – including ice-breakers, creative group projects and inter-team competitions – allowing social relationships to develop organically.

Across treatments, we randomly manipulated levels of pay inequality by varying the within- and between-group dispersion in the piece rates subjects could earn from an incentivized real-effort task. This task was repeated throughout the programme, interspersed with group activities, with each individual's pay rate fixed throughout. Incentive levels were substantial (average earnings in the real-effort task were approximately ten times the local hourly minimum wage), inequalities were made highly salient, and in the most extreme cases they amounted to the rich earning 60 times more than the poor for identical work. Group cohesion was measured at the start of the experiment and again after all rounds of the real-effort task, using the tool developed by Gächter et al. (2015, 2025), based on the Inclusion of the Other in the Self (IoS) scale popular in psychological research (Aron et al., 1992). Finally, group efficiency was assessed at the end of the study using incentivized experimental games where Pareto-efficient outcomes required successful cooperation or coordination.

Groups were assigned to one of seven different treatments. Six of these explored the effects of within-group inequality, implementing either low, medium or high levels of dispersion in the piece rates paid to members of a given group. In light of the popular argument that negative reactance to inequality is dampened when disparities result from merit rather than luck (Konow, 2003; Starmans et al., 2017; Fehr, 2018; Cappelen et al., 2023), these treatments also varied whether each individual's status within their group was determined randomly or based on performance in a preliminary task. In the seventh treatment, inspired by literature suggesting that inequalities between groups may have their own distinct effects (e.g. Hargreaves Heap et al., 2015; Levi and Ramalingam, 2025), we set within-group inequality to a low level but also introduced salient pay rate differences between each group in the same session.

As expected, we found a relationship between the development of group cohesion and interaction efficiency: groups who formed closer relationships worked more productively together. Moreover, we observed the hypothesised effect of between-group inequality: when different groups received different pay rates, this tended to enhance the development of social bonds within each group. This result has real-world implications. Even in highly unequal societies, a large proportion of interactions may take place between individuals of similar socioeconomic status. The distinctiveness between rich and poor groups may promote class solidarity, facilitating strong cohesion among those who regularly work together.

To our surprise, however, we found no support for the hypothesis that group cohesion would be stymied by within-group inequality. Groups were able to build substantial levels of cohesion,

regardless of the extremity of pay disparities between members. Furthermore, we lack strong evidence that the development of group cohesion depended on whether within-group inequalities were determined by luck or merit. Correspondingly, we found no tendency for any negative effects of inequality on efficiency.

Like any empirical study, ours was conducted in a specific setting and may generalize better to some external contexts than others (List, 2020). Our experiments were conducted with student subjects at two rather different universities in China. One was the University of Nottingham Ningbo China, a Sino-foreign institution in a wealthy city on the East Coast, whose students are generally from rich families and relatively Western-influenced. The other was Zhongnan University of Economics and Law in the central city of Wuhan, a relatively prestigious Chinese university, whose students tend to be academically high-flying but not especially rich. Given the socioeconomic differences between these universities, one might imagine that the subjects in Wuhan would care more about the money on offer in an economic experiment and be more perturbed by earnings inequalities during it. From the perspective of external validity, it is therefore somewhat reassuring that the results we obtained were, on the whole, qualitatively similar at the two universities.

Overall, we find a lack of evidence for one plausible channel – group cohesion – through which inequality may harm efficiency, and even find that when inequality arises between groups it can have positive impacts on this channel. Our results are therefore contrary to the school of thought that inequality is harmful, and are more in line with the traditional idea in economics of inequality being associated with greater efficiency (e.g. Okun, 2015). We do not claim that our study shows inequality is harmless *in general*, since by design our investigations are limited to a single channel. But our results indicate that group cohesion – an important economic and social asset – may be surprisingly resilient to inequality between individuals, and even reinforced by inequality with other groups.

Our study contributes to several strands of literature. Primarily, it adds to the extensive body of work on the relationship between equality and efficiency (e.g. Browning and Johnson, 1984; Blank, 2002; Bourguignon et al., 2007; Abramitzky, 2011; Okun, 2015; Sachs, 2017). While this area has been widely explored, identifying causal effects of equality on efficiency is extremely challenging. Recently, researchers have increasingly turned to experimental methods to address this, measuring group efficiencies using laboratory games. These studies often find within-group inequality undermines efficient interaction (e.g. Cherry et al., 2005; Anderson et

al., 2008; Hargreaves Heap et al., 2016; Gächter et al., 2017; Markussen et al., 2021), though some report the opposite (e.g. Waichman et al., 2021; Oconnor et al., 2022; Chisadza et al., 2023; for reviews of this literature and its mixed findings, see Levi and Ramalingam, 2025; Suchon and Theroude, 2025). Such research differs from ours in that inequalities are generally embedded in the game parameters, rather than emerging from prior activities (with a few exceptions, e.g. Fehr et al., 2020; Grosch and Rau, 2020). Moreover, these studies do not employ face-to-face interactions which could facilitate group cohesion as a possible channel for inequality to act through. In this respect, the field evidence from Breza et al. (2018) is more relevant. Their study finds experimentally induced pay disparities between individual manufacturing workers reduce productivity (though the effect disappears when it becomes salient that the inequalities are merit-based). A breakdown in group cohesion is discussed as a possible channel for the effect, but their measure of cohesion is only indirect, relying on cooperation in experimental games.[1]

The idea that inequality may damage social cohesion has frequently been argued, supported by evidence linking inequality to negative societal outcomes (e.g. Alesina and Perotti, 1996; Alesina and La Ferrara, 2002; Wilkinson and Pickett, 2010; Olaoye et al., 2025). Again, however, obtaining direct causal evidence is very difficult. Our study is one of very few employing a causal identification design to estimate the effects of inequality on social relationships using IoS measurements.

A recent paper by Stancato et al. (2024) used a deceptive design to measure interpersonal closeness in dyadic relationships, finding that increased pay inequality reduced the worse-off individual's sense of closeness towards their partner; as in our study, this outcome was unaffected by whether the inequality resulted from luck or merit. In a study somewhat similar to ours, Ranehill et al. (2023) used post-task redistribution to manipulate levels of within-group inequality following a real-effort task. They then measured group cohesion, as well as levels of cooperation in experimental games. Most outcomes were unaffected by the inequality manipulation, but group cohesion was improved by equality-enhancing redistribution. While both of these studies contrast with our findings by reporting negative effects of within-group

[1] Other lab and field experiments which manipulate pay inequalities in the piece rates offered for performing real-effort tasks produce mixed evidence on the effects on productivity. See the review in Fongoni (2024).

inequality on social ties, it is important to note both employed levels of social interaction that were minimal in comparison to our design, with no in-person contact.

While the bulk of the literature we cite focuses purely on inequalities that exist within groups, a key contribution of our study is to differentiate between the effects of within-group and between-group inequalities. Existing studies on between-group inequality have found its effects to be detrimental, in reducing public goods contributions (Brañas-Garza et al., 2021; Levi and Ramalingam, 2025), increasing inter-group conflict (Fallucchi et al., 2021; Eckel et al., 2022; Baier et al., 2024) and triggering instability (Barr et al., 2024).[2] Our finding presents a counterpoint to these negative effects, demonstrating that inequality between groups can also have the positive consequence of strengthening cohesion within groups. This represents a mechanism previously overlooked in the discourse on inequality and efficiency, and refutes the prevailing argument that the effects of inequality on cohesion must necessarily be negative.

Finally, we contribute to the literature on the relationship between group cohesion and productivity (e.g. Casey-Campbell and Martens, 2009; Evans and Dion, 1991; Forsyth, 2021; Sidorenkov and Borokhovski, 2023). Gächter et al. (2025) demonstrate that group cohesion is a powerful predictor of group efficiency in weak-link games; we show that this extends to efficiency in public goods games. In their study, cohesion is measured among groups of pre-existing friends, while in ours we allow it to develop organically among groups of strangers. While neither their study nor ours is set up to identify causal effects of group cohesion, our observation that groups with more rapidly developing cohesion are also able to cooperate more efficiently lends further credence to Gächter et al.'s argument that their tool for measuring group cohesion represents a useful summary statistic capturing features of real groups which affect team production.

2. Methods

2.1. Experimental design

The key focus of our experiment was to investigate whether inequalities in earnings opportunity would influence the development of group cohesion and subsequent efficiency of within-group

[2] More nuanced are the findings of Hargreaves Heap et al. (2015), which suggest between-group inequalities may either increase or decrease team effort, depending on the extent of inequality and whether it advantages or disadvantages the team.

economic interactions. In order to allow meaningful evolution of social relationships, we implemented a non-standard experiment, with each session conducted in a university classroom, consisting of extensive face-to-face interaction between participants, and lasting for about half a day.

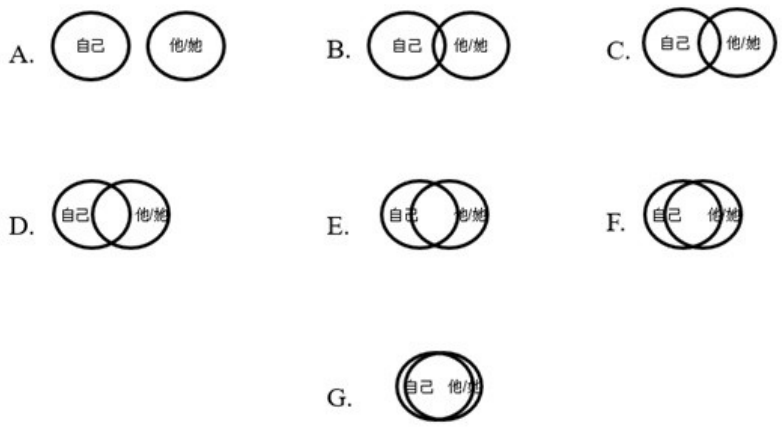
The experiment comprised two phases. In Phase One we first randomly divided experimental subjects into groups of four and measured the initial level of group cohesion among each group's members. Subjects then performed a real-effort task under inequality in the pay rates, as well as a range of team-building activities. In order to increase the salience of the inequality, the real-effort task was performed for three sets of five rounds. At the end of this phase we measured again group cohesion, in order to observe how this had developed during the experiment. In Phase Two the subjects played experimental games where group success depended on cooperation or coordination, allowing us to measure efficiency of interactions.

Our experimental treatments introduced both different degrees of inequality in the real-effort task and different procedures for assigning this inequality among subjects. Inequalities were either determined by luck or merit. We will study whether these treatments influenced the development of group cohesion and the behaviour in the subsequent experimental games, and check for mediating effects of group cohesion on efficient behaviour. The following paragraphs discuss the two phases of the experiments and then explain in detail the experimental treatments as well as the procedures.

Phase One

At the start of the experiment the subjects were randomly divided into groups of four and each group was seated at separate tables. The subjects would stay in the same group throughout the session. The first part of Phase One was a questionnaire, which included questions enabling us to obtain a measure of group cohesion. Using the IoS scale, each subject reported his or her closeness to each of the other three group members. This was recorded on a scale from 1 to 7, representing the degree to which the selected pair of circles overlapped (see Figure 1). Crucially, the responses were collected privately, and subjects were informed that these ratings would never be revealed to other group members. We follow Gächter et al. (2025), constructing an index of group cohesion by taking the lowest closeness each subject reports towards any fellow group member, and calculating the average of these scores among the four respondents in the group.

Figure 1: Inclusion of the Other in the Self scale

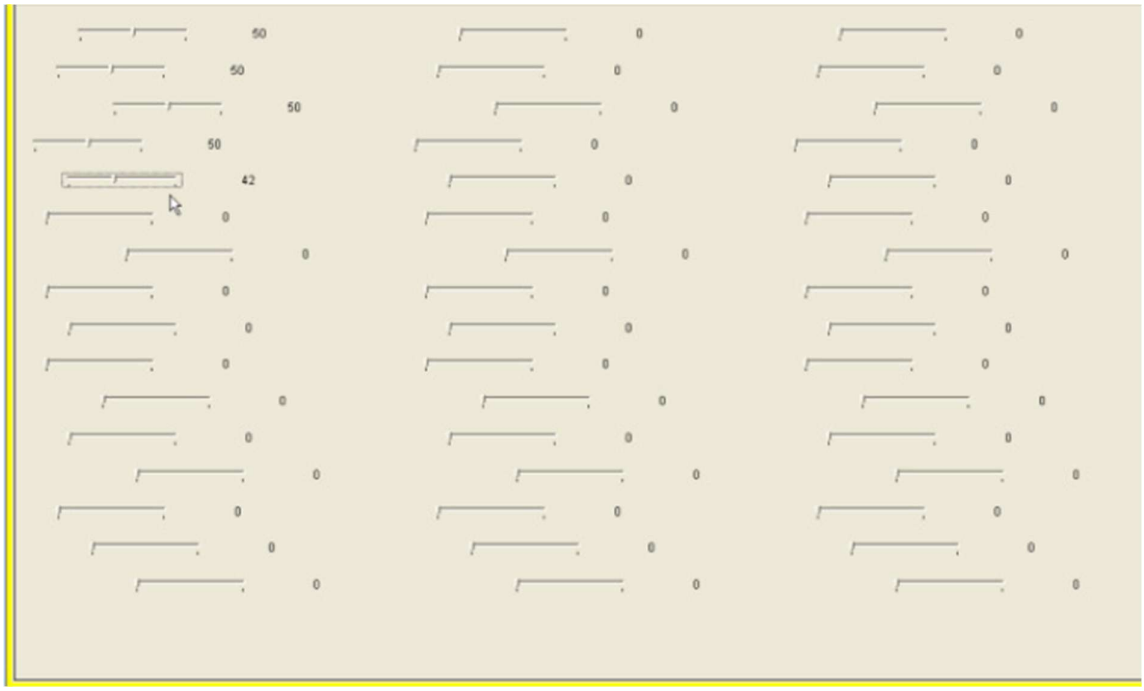


Notes: The scale is adapted from Aron et al. (1992) and translated into Chinese.[3] The labels in the left-hand circles in Figure 1 read 'self' and those in the right-hand circles read 'other'. Response A = 1 corresponds to the lowest possible level of connection with a given group member (no overlap of the two circles). Subsequent letters indicate a higher degree of connection, with response G = 7 implying the closest level of connection.

Following the completion of the first questionnaire, the subjects performed the first set of rounds of the real-effort task. All subjects were given tablets to perform for five two-minute rounds the slider task introduced by Gill and Prowse (2019). The task consisted of using the mouse to position sliders initially placed at integer location 0 to the value of 50. Each slider had a number to its right showing its current position (Figure 2 presents how the task was displayed to the participants). There was a total of 48 sliders in each round, and each slider correctly positioned at exactly 50 would earn a subject one point. To familiarise subjects with the slider task, there were three practice rounds prior to this first set of incentivized rounds.

[3] The IoS scale has previously been translated into Chinese and used among Chinese samples, for instance by Meng et al. (2022) and Xie et al. (2024).

Figure 2: Slider Task as displayed to the subjects (example of round in progress)



The crucial element of this activity was that while all team members performed the same task, the rate of pay differed across the members. Before the start of the task, the subjects were assigned to one of the four roles, which determined the specific pay rates. Depending on the role, different subjects within the same group would earn different amounts of money for every point from the task.

After the subjects completed the slider task, the experiment proceeded with two unincentivised team building tasks. First, the subjects were given a series of ice-breaker questions about themselves, which they needed to discuss in their group for about 10 minutes. These were aimed at developing subjects' knowledge about their team members' backgrounds, interests, opinions, life ambitions and so forth. Subsequently, the subjects were given an exercise in which they were asked to design a new mobile phone app. They needed to discuss the idea within their groups for 10 minutes and then give a two-minute presentation to the other groups in the room.

The subjects then performed again the slider task for another five two-minute rounds. Importantly, the pay rates determined initially remained unchanged, as the subjects within each group kept the same roles as before. This was followed by a 30-minute meal break. The meal was delivered to each group's table and team members ate in each other's company.

In the subsequent part of the experiment, the subjects performed the same real-effort task again, in the same roles, for a final set of five two-minute rounds, which was followed by a final, 30-minute team-building task. This consisted of a competition between the different groups in the room. The experimenter asked quiz questions that the group had to discuss and answer collectively. There were no incentives for this part of the experiment but the teams' performances were written on the board.

In the next stage of the experiment, subjects took another questionnaire, which included the second measurement of the IoS items, enabling us to observe how group cohesion had changed relative to the measurement at the start of the session.

Phase Two

In Phase Two the subjects played a series of experimental games. The order of these was randomised across sessions, and subjects were told that only one of these would later be randomly selected for the payment for this phase of the experiment. The focus of the current paper is on the two games outlined below.[4]

Public goods game

In this task subjects played the following game for eight rounds, always matched into the same four-member groups as in Phase One.[5] In every period they were each endowed with 8 RMB[6] and had to decide how much of this endowment to keep for themselves and how much to

[4] In addition to these two games, subjects also undertook a donation task, consisting of dictator games. These do not relate to the research questions that we focus on in this paper. The task required subjects to make three decisions. In each decision problem the subjects were given 160 RMB and had to decide what proportion of this initial endowment to donate to a specific charity. The only difference across these decision problems was whether the charity was local, national or international. We will not use the data from these decisions in the empirical analysis for this paper.

[5] Public goods games are a standard tool in experimental economics to study cooperative behaviour (e.g. Andreoni, 1988; Fischbacher and Gächter, 2010).

[6] At the time of the experiment, 1 RMB was nominally worth approximately 0.15 USD.

contribute to the group project. Each RMB contributed to this public good would increase the earnings of every group member by 0.4. The earnings of each subject consisted of the amount kept to himself/herself and his/her returns from the group project, where the total sum of contributions in the group would be multiplied by 1.6 and then divided equally between all members.

The payoff in RMB to player i for each round is given by the following equation.

$$\pi_i = 8 - c_i + 0.4 \cdot \left(c_i + \sum_{j=1}^{3} c_{j \neq i} \right)$$

where c_i denotes the amount of RMB contributed to the public good by group member i.

This game introduces the tension between personal and collective gains. Under the assumption of profit-maximizing selfish individuals, each subject would keep all the endowment to himself/herself. However, the payoffs of all individuals would be maximised if all contributed their total endowments to the public good. As such, the game resembles common real-world team production problems, where efficient outcomes require overcoming such social dilemmas.

Coordination game

In this task subjects in the same four-member groups played for eight rounds a weak-link coordination game. Our version of the game is the same as in Gächter et al. (2025), whereby all of the four group members simultaneously choose an integer between 1 and 5 as their effort levels. Payoffs to each subject were determined both by his/her chosen level of effort and by the lowest level of effort among all the group members.

The payoff in RMB to player i for each round is given by the following equation.

$$\pi_i = 14 - 4e_i + 6 * [\min(e_1, \dots, e_4)]$$

where e_i is the effort level chosen by group member i. This payoff function implies that the revenue for each group member is determined solely by the effort level of the group member providing the lowest effort, but each individual's costs increase in response to an increase in their own effort level, even if these are above the minimum level in the group. The payoff function generates the payoff matrix in Table 1, which was shown to subjects.

Table 1: Earnings table in weak-link game (per round)

	The smallest value of effort chosen					
		1	**2**	**3**	**4**	**5**
	1	16				
	2	12	18			
Your effort	**3**	8	14	20		
choice	**4**	4	10	16	22	
	5	0	6	12	18	24

Given that effort in this game is costly and any effort above that of the lowest-providing player is wasted, each subject's payoff-maximizing behaviour is to choose the effort level which matches that of the lowest of the other group members. Efficient outcomes in the game require a mixture of coordination (i.e. everyone choosing the same effort level) and cooperation (i.e. everyone choosing high effort levels, despite the risk of such effort being wasted). Our game has five strict Pareto-ranked Nash equilibria, whereby a choice of the highest level of effort by all four group members would lead to the unique social optimum. Despite the existence of a Pareto-dominant Nash equilibrium, efficient outcomes are in practice difficult to achieve in this type of game (Van Huyck et al., 1990), with effort levels tending to decline when it is played repeatedly. Gächter et al. (2025) found effort levels in the game could be predicted by a group's level of measured cohesion.

2.2. Experimental treatments and hypotheses

Our first treatment manipulation was the degree of within-group inequality built into the payoff schemes for the real-effort task. The second manipulation was whether the inequality in pay rates within a group was determined by randomisation or by tournament, namely, by the performance of group members in a preliminary real-effort task. In the sessions where pay schemes were assigned by tournament, before the first incentivised period of the slider task the subjects within each group competed for roles in a qualifying round, which took place after the three practice rounds. They were ranked according to their scores in this round, and assigned roles in accordance to this ranking, with the best-performing member obtaining the highest pay rate and the worst-performing member the lowest. Subjects were informed in advance that the qualifying round would be used for this purpose. In addition to the within-group inequality

manipulations, we also have a treatment manipulating variation in the pay rates between groups.

There were a total of seven treatments. Three treatments were designed to explore the effects of levels of within-group inequality determined by randomisation (*Low-inequality Baseline*, *Medium-inequality Baseline*, *High-inequality Baseline*). A further three implemented analogous levels of within-group inequality but determined by initial performance (*Low-inequality Merit-based*, *Medium-inequality Merit-based*, *High-inequality Merit-based*). The *Between-group-inequality* treatment implemented a low degree of within-group inequality based on randomisation – equivalent to the *Low-inequality Baseline* treatment – but also introduced inequality between groups, also determined by randomisation. Specifically, there were four randomly assigned pay levels, with the highest earnings (Pay Level 1) assigned to one group, and progressively lower earnings assigned to the subsequent three groups. In all treatments, the pay rate assigned to a given subject – and in the *Between-group-inequality* treatment, the pay level assigned to their group – remained constant throughout all rounds of the slider task.

Table 2: Experimental treatments

Treatment	Slider Task: Earnings per point (RMB)				Gini (within-group)
	Role 1	Role 2	Role 3	Role 4	
Low-inequality Baseline	0.45	0.44	0.43	0.43	0.01
Medium-inequality Baseline	0.88	0.44	0.29	0.15	0.33
High-inequality Baseline	1.21	0.44	0.09	0.02	0.56
Low-inequality Merit-based	0.45	0.44	0.43	0.43	0.01
Medium-inequality Merit-based	0.88	0.44	0.29	0.15	0.33
High-inequality Merit-based	1.21	0.44	0.09	0.02	0.56
Between-group-inequality	Earnings vary across Pay Levels*				0.01

** Pay level 1 - Role 1: 0.9; Role 2: 0.88; Role 3: 0.86; Role 4: 0.86; Pay level 2 - Role 1: 0.45; Role 2: 0.44; Role 3: 0.43; Role 4: 0.43; Pay level 3 - Role 1: 0.3; Role 2: 0.29; Role 3: 0.29; Role 4: 0.29; Pay level 4 - Role 1: 0.15; Role 2: 0.15; Role 3: 0.14; Role 4: 0.14.*

Table 2 summarises all experimental treatments. The degree of within-group inequality in earning opportunities, measured by the Gini coefficient, follow from the pay rates for Roles 1, 2, 3, and 4. In the low within-group inequality treatments, pay rates are almost identical across group members, reflected by the very low Gini of 0.01. In contrast, the dispersion in pay rates

in the high within-group inequality treatments is extreme, with the best role earning more than 60 times the piece rate of the worst role, amounting to a Gini of 0.56. In the *Between-group-inequality* treatment, the within-group Gini is 0.01, while the Gini for pay rates measured between groups is 0.33, equivalent to the within-group inequality Gini in the *Medium-inequality treatments*.

Motivated by the arguments put forward in the introduction and by the emerging research on inequality, cohesion and economic efficiency, we designed our experimental treatments to test the following five research hypotheses.

Hypothesis 1: *The development of group cohesion is negatively affected by within-group inequality.*

We test this hypothesis by comparing how group cohesion develops with different levels of within-group inequality, both within the *Baseline* and *Merit-based* treatments. Specifically, we hypothesise development of group cohesion will be stronger under the *Medium-inequality Baseline* treatment than under the *High-inequality Baseline*, and stronger still under the *Low-inequality Baseline*. We also hypothesise development of group cohesion will be stronger under the *Medium-inequality Merit-Based* treatment than under the *High-inequality Merit-Based* treatment, and stronger still under the *Low-inequality Merit-Based* treatment.

Hypothesis 2: *The development of group cohesion is positively affected by between-group inequality.*

To test this hypothesis we compare how group cohesion develops in the *Between-group-inequality* treatment relative to the *Low-inequality Baseline*.

Hypothesis 3: *The development of group cohesion is stronger if inequality is determined by merit as opposed to by luck.*

This hypothesis is tested by comparing how group cohesion develops in each of the *Low-inequality*, *Medium-inequality* and *High-inequality Baseline* treatments compared to the *Merit-based* treatment with the equivalent level of inequality.

Hypothesis 4(a & b): *Stronger development of group cohesion is associated with more efficient game outcomes.*

We address this by testing the relationships between development of group cohesion and (a) cooperation, as measured by average contributions in the public goods game; (b) coordination efficiency, as measured by average payoffs in the weak-link game.

Hypothesis 5(a & b): *Higher within-group inequality results in less efficient outcomes.*

We test this hypothesis by comparing the game outcomes with different levels of within-group inequality, both within the *Baseline* and *Merit-based* treatments. Specifically, we hypothesise (a) cooperation in the public goods game and (b) coordination efficiency in the weak-link game will be higher under the *Medium-inequality Baseline* treatment than under the *High-inequality Baseline*, and higher still under the *Low-inequality Baseline*. We also hypothesise (a) cooperation in the public goods game and (b) coordination efficiency in the weak-link game will be higher under the *Medium-inequality Merit-based* treatment than under the *High-inequality Merit-based* treatment, and higher still under the *Low-inequality Merit-based* treatment.

Hypothesis 6(a & b): *Higher between-group inequality results in more efficient game outcomes.*

Specifically, we hypothesise (a) cooperation in the public goods game and (b) coordination efficiency in the weak-link game will be higher under the *Between-group-inequality* treatment than under the *Low-inequality Baseline*.

Hypothesis 7(a & b): *Merit-based within-group inequality results in more efficient game outcomes than luck-based within-group inequality.*

Specifically, we hypothesise (a) cooperation in the public goods game and (b) coordination efficiency in the weak-link game will be higher under each *Merit-based* treatment than the *Baseline* treatment with the equivalent level of inequality.

Note that Hypotheses 5-7 are based on Hypotheses 1-3 and Hypothesis 4. That is, the hypothesised treatment effects on outcome variables in Hypotheses 5-7 are predicated on the corresponding treatment effects on development of group cohesion in Hypotheses 1-3, with the expectations that these effects will be mediated by the relationship between group cohesion and efficiency laid out in Hypothesis 4.

2.3. Procedures[7]

Our experiment was conducted in Mandarin Chinese at two Chinese universities: the University of Nottingham Ningbo China (in Ningbo) and Zhongnan University of Economics and Law (in Wuhan). All sessions were conducted between April and June 2022. Student participants were recruited using the ORSEE software (Greiner, 2015) and the experiment was computerised and programmed with z-Tree (Fischbacher, 2007).

Of the 49 sessions, 14 were conducted in Ningbo and 35 in Wuhan. We aimed for 16 subjects per session but in some cases went ahead with sessions with fewer, due to no-shows. However, the *Between-group-inequality* sessions only went ahead with exactly 16 subjects, as the design required four groups with pay levels varying between them. Table 3 reports the background of the sample, and the total number of subjects in each treatment, at each university. Demographic characteristics are similar between the Ningbo and Wuhan samples. Subjects were 72% female, and 21 years old on average. However, the two samples differ by social background. Three-fifths of subjects in Wuhan were born outside of cities, while only one-tenth in Ningbo were. As the subjects were drawn from different academic disciplines at large universities, cases of pre-existing relationships between group members were likely to be rare, but were possible; it is therefore important that we measured group cohesion at the start of the experiment to account for this.

At the start of each experimental session the subjects were randomly assigned to the groups of four and each group would be seated at a separate table throughout the experiment. Each experimental subject was provided with a tablet. Subsequently, the main experimenter read aloud an introduction and instructions, which were also distributed in paper format to each

[7] Prior to running the experiment reported here, we ran three pilot sessions in 2021 which consisted only of Phase One of the experiment, all using the *Low-Inequality Baseline* treatment. The sole purpose of this pilot was to establish that the activities in Phase One were sufficient to generate substantial development of group cohesion, as identified by IoS ratings differing between their first and second measurements. This was a pre-requisite for our experiment being equipped to test whether any such development of group cohesion would be affected by treatment variations. The pilot confirmed that substantial development occurred in this fixed setting. Since it did not measure game outcomes and was not designed to test any of the paper's hypotheses, its data are not used in this paper's analyses.

subject. The experimental instructions are presented in Appendix A (English translation) and Appendix B (original Chinese version).

Table 3: Descriptive Statistics of participants by location

	Total		Ningbo		Wuhan	
Age	21.3	(2.1)	20.8	(1.7)	21.6	(2.3)
Gender						
Male	206	28.0%	58	28.4%	148	27.8%
Female	530	72.0%	146	71.6%	384	72.2%
Place of Birth						
Municipality/Provincial Capital	136	18.5%	67	32.8%	69	13.0%
Small/Medium City	258	35.1%	115	56.4%	143	26.9%
Town/Village	340	46.2%	20	9.8%	320	60.2%
Outside of China	2	0.3%	2	1.0%	0	0.0%
Major						
Philosophy	17	2.3%	4	2.0%	13	2.4%
Economics	307	41.7%	100	49.0%	207	38.9%
Law	121	16.4%	1	0.5%	120	22.6%
Education	5	0.7%	5	2.5%	0	0.0%
Literature	75	10.2%	14	6.9%	61	11.5%
History	8	1.1%	3	1.5%	5	0.9%
Science	39	5.3%	24	11.8%	15	2.8%
Engineering	41	5.6%	28	13.7%	13	2.4%
Agriculture	2	0.3%	0	0.0%	2	0.4%
Management	114	15.5%	24	11.8%	90	16.9%
Fine Art	7	1.0%	1	0.5%	6	1.1%
Treatment						
Low-inequality Baseline	104	14.1%	32	15.7%	72	13.5%
Medium-inequality Baseline	100	13.6%	24	11.8%	76	14.3%
High-inequality Baseline	104	14.1%	28	13.7%	76	14.3%
Low-inequality Merit-based	104	14.1%	28	13.7%	76	14.3%
Medium-inequality Merit-based	104	14.1%	28	13.7%	76	14.3%
High-inequality Merit-based	108	14.7%	32	15.7%	76	14.3%
Between-group-inequality	112	15.2%	32	15.7%	80	15.0%
Obs.	736		204		532	

Note: Age is a continuous variable and thus the mean is presented, with standard deviation in parentheses.

Group members could interact, see and communicate with each other throughout all team-building activities, as well as during the 30-minute meal delivered to each group's table. However, privacy was essential in collecting data from the questionnaires (including the recording of the group cohesion measure) and the experimental games in Phase Two. To this end, during these parts of the experiment removable separators were placed on the tables, preventing the subjects from seeing each other's responses on the tablets where they entered responses for these tasks. The separators were also erected during the slider task, to ensure that subjects worked alone on the task.

All subjects received a show-up fee of 20 RMB, plus their cumulative earnings from the slider task in Phase One and their earnings from the experimental games in Phase Two. Subjects knew from the outset that their earnings would consist of separate payments from the two phases, but they received no information about the nature of each experimental activity until they reached it; they were only told that their roles and outcomes in Phase One would have no bearing on the nature of the tasks in Phase Two. Feedback was provided to subjects during the parts of the experiment where money was at stake. After each round of the slider task, the subjects would receive feedback on the tablet screen reporting their performance and converted earnings, as well as of the other group members. At the end of the first set of slider tasks, the experimenter would announce the earnings of each subject by writing them on the board in the front of the room and reading these out. The reported earnings in the two subsequent sets of slider tasks were added to those written for the first set and were visible to all subjects throughout the experiment.

At the end of each round during the public goods game, subjects saw on their screen information about their own contribution to the project and their own earnings both for this round and accumulated for all completed periods, as well as about the total contributions in the given round by all group members. At the end of each round of the weak-link game, each subject's screen reported his/her level of effort, as well as the lowest level of effort in the group (but not the identity of the player who committed this), and his/her own earnings (both for this round and accumulated for all completed rounds). After the final round in each of these tasks, a summary screen reported subjects' task earnings. Experimental sessions lasted about three and a half hours, including the 30-minute meal break. The average total experimental earnings (comprising the show-up fee and the earnings from Phase One and Phase Two) were 262.59 RMB (38.44 USD at the time of the experiment), which equates to a relatively high hourly reimbursement for a lab experiment in China. These earnings were paid privately.

3. Results

First, we consider the extent of inequality actually generated by our treatments in Phase One. Table 4 displays the average number of sliders completed, and resultant earnings by subjects in each role (and, for *Between-group-inequality*, at each pay level) in each treatment. This table also shows the average within-group Gini in individual task earnings – and the between-group Gini in group earnings – in each treatment.

Table 4: Slider Task Outcomes

Treatment	Mean Outcomes				Gini (within-group)	Gini (between-group)
	Number of sliders completed					
	Role 1	Role 2	Role 3	Role 4		
Low-inequality Baseline	261.54	270.19	280.58	266.46		
Medium-inequality Baseline	284.96	272.68	279.92	256.64		
High-inequality Baseline	274.81	261.96	244.85	238.19		
Low-inequality Merit-based	315.85	282.73	266.08	235.46		
Medium-inequality Merit-based	323.54	296.42	271.96	213.69		
High-inequality Merit-based	314.63	278.00	255.81	208.96		
Between-group-inequality	267.89	275.43	263.61	262.89		
	Pay Level 1	Pay Level 2	Pay Level 3	Pay Level 4		
Between-group-inequality	263.04	269.04	268.39	269.36		
	Resultant Earnings (RMB)					
	Role 1	Role 2	Role 3	Role 4		
Low-inequality Baseline	117.69	118.88	120.65	114.58	0.11	0.05
Medium-inequality Baseline	250.76	119.98	81.18	38.50	0.37	0.07
High-inequality Baseline	332.52	115.26	22.04	4.76	0.60	0.10
Low-inequality Merit-based	142.13	124.40	114.41	101.25	0.11	0.04
Medium-inequality Merit-based	284.71	130.43	78.87	32.05	0.41	0.07
High-inequality Merit-based	380.70	122.32	23.02	4.18	0.60	0.07
Between-group-inequality	121.47	120.28	115.73	108.05	0.12	
	Pay Level 1	Pay Level 2	Pay Level 3	Pay Level 4		
Between-group-inequality	230.29	117.69	78.49	39.06		0.34

This shows that slider task outcomes emerged as expected. Roles with higher pay rates tended to complete more sliders, especially in the merit-based treatments where roles were endogenous to task ability. However, clearly the key determinant of within-group inequality in earnings was the degree of within-group inequality in earning opportunities; across treatments, the Gini for

this outcome variable closely reflected the within-group Gini for pay rates, with the highest levels in both *High-inequality* treatments. In the *Between-group inequality* treatment, there were small differences in average earnings between roles but larger differences between groups at different pay levels, resulting in a between-group Gini of 0.34. The corresponding values for this variable in all other treatments were low, ranging from 0.05 to 0.1. Taken together, this data shows the treatment manipulations were successful in generating inequalities as intended.[8]

Group Cohesion

Table 5 reports the average group cohesion in each treatment at the beginning and end of Phase One. The initial levels are close to the minimum, reflecting the fact that most subjects were strangers. Throughout this phase of the experiment, group cohesion increased substantially – by more than one scale-point – in all treatments. However, this development was not identical across treatments. It was highest in the *Between-group-inequality* treatment.

Hypothesis 1 contended that the change in group cohesion would be more positive for low within-group inequality treatments than for corresponding treatments with medium and high within-group inequality. However, Table 5 shows that this variable in fact takes a lower value in the *Low-inequality Baseline* than in either the *Medium-inequality* or *High-inequality Baseline* treatments. Furthermore, it takes a lower value in the *Low-inequality Merit-based* treatment than in the *High-inequality Merit-based* treatment. Significance of treatment differences is assessed using Wilcoxon rank-sum tests. Contrary to our hypothesis, the change in group cohesion is significantly higher in the *Medium-inequality Baseline* than the *Low-inequality Baseline* ($p=0.022$). There are no significant differences between the *Low-inequality Baseline* and the *High-inequality Baseline*, or between the *Medium-inequality Baseline* and the

[8] While productivity in the slider task differs across roles, differences across treatments in the average number of sliders groups complete appear minimal, and regression analysis confirms they are not significant. While not the focus of our investigations, we therefore note that our study obtains no evidence for efficiency losses resulting from experimentally assigned inequalities in terms of real-effort task output. This contributes to the mixed evidence on this topic, reviewed in Fongoni (2024).

High-inequality Baseline. Meanwhile, corresponding pairwise tests for the *Merit-based* treatments find no significant differences.[9] Therefore, Hypothesis 1 is not supported.

For Hypothesis 2, we compare the change in group cohesion between the *Between-group-inequality* treatment and the *Low-inequality Baseline*. This Wilcoxon rank-sum test finds the difference is significant ($p<0.001$). This supports our hypothesis that between-group inequality strengthens the development of group cohesion.

Table 5: Levels of group cohesion before and after the slider task by treatment

	Treatment							
	Baseline Low	Baseline Medium	Baseline High	Between Group	Merit Low	Merit Medium	Merit High	Total
Pre-task	1.404	1.350	1.462	1.348	1.413	1.404	1.528	1.416
	(0.495)	(0.382)	(0.532)	(0.362)	(0.583)	(0.442)	(0.547)	(0.479)
Post-task	2.654	2.970	2.827	3.071	2.779	2.683	3.093	2.871
	(0.592)	(0.758)	(0.706)	(0.597)	(0.606)	(0.611)	(0.794)	(0.680)
Change	1.250	1.620	1.365	1.723	1.365	1.279	1.565	1.455
	(0.469)	(0.743)	(0.668)	(0.515)	(0.420)	(0.481)	(0.566)	(0.577)

Note: Standard deviations in parentheses.

For Hypothesis 3, we compare the change in group cohesion between each *Baseline* treatment and its equivalent *Merit-based* treatment. Under low within-group inequality, group cohesion developed significantly more strongly when inequality was merit-based ($p=0.049$). The treatment effect is in the same direction but not significant for high within-group inequality, while it is in the opposite direction and insignificant for medium within-group inequality.

[9] The Wilcoxon rank-sum tests used in this section of the paper treat each group as providing one independent observation. Their results are reported in full in Appendix C (Table C2). All tests in this paper are two-sided.

Overall, there is not consistent evidence to support the hypothesis that group cohesion would develop more strongly when within-group inequalities are merit-based.[10]

These non-parametric tests are backed up by OLS regression analysis, with the change in group cohesion as the dependent variable, regressed against treatment dummies. In order to include demographic control variables, observations are at the subject level, but as the dependent variable is the same for all members of the same group we cluster the standard errors at the group level.[11] Output is presented in Table 6. The key result of stronger development of group cohesion under between-group inequality than under the corresponding baseline (the omitted treatment category) is clearly supported. No support is found for either Hypothesis 1 or Hypothesis 3; in contrast to the rank-sum test, this analysis finds no significant difference between the *Low-inequality Baseline* and *Low-inequality Merit-based* treatments. Models (2) and (4) control the levels of income subjects earn from the slider task, as this could in principle represent a channel through which any treatment effects on group cohesion operate; however, this appears not to be the case, as including this control makes very little difference. In models (3) and (4) we also control for the proportion of female members in the group. We included

[10] Our construction of the group cohesion statistic follows the approach introduced by Gächter et al., 2025. In Appendix C, we provide further analyses that adopt alternative approaches to calculate group cohesion. Alternative 1 takes the lowest reported closeness each group member *receives* from the other three, and averages this within each group. Alternative 2 calculates the group's cohesion as the lowest of all 12 elicited closeness ratings, while Alternative 3 calculates it as the average of them. Redoing our analyses using these alternative measures generates qualitatively similar results, although the treatment difference in development of group cohesion between *Low-inequality Baseline* and *Between-group-inequality* is significant only at the 10% level using Alternative 2 (it remains strongly significant using the other alternatives). This is not surprising as Alternative 2 is a cruder way of measuring group cohesion – throwing away all but one observation per group – and we would argue conceptually weaker than our preferred measure. Using these alternative measures, the differences between *Low-inequality Baseline* and *Medium-inequality Baseline*, and between *Low-inequality Baseline* and *Low-inequality Merit-based* lose significance.

[11] There is unresolved debate over whether, in experiments with data structure like ours, it is preferable to cluster standard errors at the group or session level. The arguments of Kim (2020) would support our approach of group-level clustering. In our case, however, the output looks almost identical – and conclusions about treatment effects are unchanged – when we instead cluster at the session level, with observations at either the individual or group level (available on request).

this variable in response to suggestions from seminar audiences that the gender composition of the group might be an important determinant of group cohesion, but it is not significant and including it has very little impact on the treatment effects.[12]

To delve deeper into the hypothesised effects of within-group inequality that fail to appear, we analyse the relationships that develop between particular roles in the groups. If inequality were harmful, we would expect this to be reflected by relatively weak relationships between the group members (Roles 1 and 4) with the greatest pay disparities. However, Table D1 (Appendix D), which reports the change in closeness towards other group members reported by those in each role in each treatment, reveals no obvious relationship in any treatment between the dyadic relationship and how effectively social closeness was able to develop. There was also no general tendency for higher closeness scores to be reported by or towards relatively rich or poor group members.

In a similar vein, Table D2 (Appendix D) explores how the development of group cohesion varies between groups at different pay levels in the *Between-group-inequality* treatment. For our preferred measure of group cohesion, there are no significant differences between the groups at different pay levels, suggesting that the strong development of cohesion in this treatment was not driven particularly by either the rich or the poor.[13]

[12] We also ran a further model (not presented) with interaction terms between the treatment dummies and the female proportion, which were found not to be significant. This indicates that the induced inequalities did not have divergent effects on male- and female-dominated groups.

[13] Relatedly, readers may wonder whether the *Between-group-inequality* treatment, in addition to impacting the average development of group cohesion, also led to more dispersion in this variable across groups, since different groups were affected differently by this treatment. Figure E1 in Appendix E plots the distribution of group cohesion change in each treatment. While there was indeed more dispersion in this treatment than in the corresponding *Baseline*, other treatments saw even greater dispersion.

Table 6: OLS Regressions on Change in Group Cohesion

	(1)	(2)	(3)	(4)
Treatment:				
Baseline Low	ref.	ref.	ref.	ref.
	(.)	(.)	(.)	(.)
Baseline Medium	0.368*	0.368*	0.369*	0.369*
	(2.19)	(2.19)	(2.21)	(2.21)
Baseline High	0.117	0.117	0.115	0.115
	(0.74)	(0.74)	(0.73)	(0.73)
Between Group	0.482***	0.482***	0.482***	0.481***
	(3.64)	(3.64)	(3.60)	(3.59)
Merit Low	0.126	0.126	0.126	0.126
	(1.04)	(1.04)	(1.04)	(1.04)
Merit Medium	0.0342	0.0346	0.0363	0.0369
	(0.26)	(0.27)	(0.28)	(0.29)
Merit High	0.312*	0.312*	0.311*	0.311*
	(2.24)	(2.25)	(2.24)	(2.24)
Controls:				
Age	0.0260*	0.0260*	0.0259*	0.0259*
	(2.16)	(2.15)	(2.15)	(2.14)
Gender	0.0131	0.0121	0.00615	0.00475
	(0.27)	(0.25)	(1.00)	(0.56)
Slider Income		-0.0000367		-0.0000404
		(-0.22)		(-0.24)
Female proportion			0.0269	0.0282
			(0.14)	(0.15)
Constant	0.682*	0.688*	0.670*	0.676*
	(2.52)	(2.50)	(2.30)	(2.30)
N	734	734	734	734

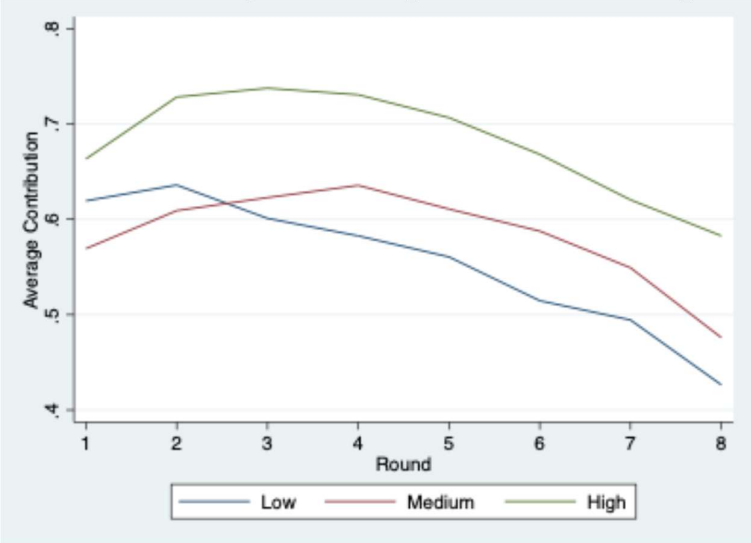
Notes: The dependent variable is the change in group cohesion. Standard errors are clustered at the group level. t statistics in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Game Outcomes

To assess the role of group cohesion in the experimental games, we partition the groups into three categories, according to their change in group cohesion during Phase One. Groups are defined as having low cohesion development if the change in the group cohesion statistic is less than +1 scale-points, as having medium cohesion development if the change is between +1 and +1.25, and high cohesion development if the change is greater than +1.25.

Figure 3 plots average contributions in the public goods game over its eight rounds, pooling all treatments. The typical decline in contributions is observed, although contribution levels were relatively high throughout, which may reflect the game being played between people known to one another (see e.g. Andreoni and Petrie, 2004). However, it is apparent that cooperation was highest throughout in the groups categorised as having the highest development of cohesion, and these groups thus achieved higher efficiency (which in this game is in direct proportion to contribution levels).

Figure 3: Group contribution in the public goods game

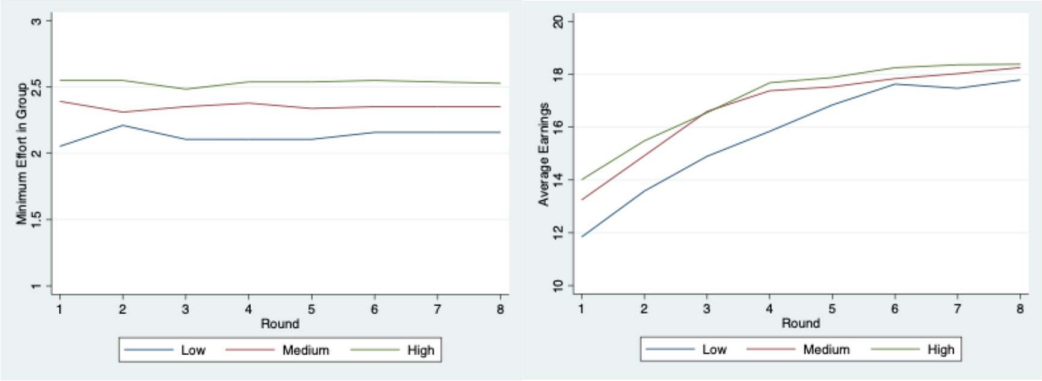


Note: In the low partition (19 groups), change in group cohesion < 1; in the medium partition (74 groups), change in group cohesion is between 1 and 1.25; in the high partition (91 groups), change in group cohesion > 1.25. Average contribution is reported as a proportion of the endowment.

Figure 4 presents a similar analysis for the weak-link game. In this game payoffs depend to a large extent on the minimum effort level within one's group. The left-hand panel of the figure presents how this varied across the different group categories, while the right-hand panel displays mean earnings in the task, throughout the game's rounds. It is notable that the

minimum effort levels in this game did not decline over time as is often observed in this game (Van Huyck et al., 1990); the earnings achieved therefore increased over time and reached high levels. Again, this atypical pattern of behaviour may be due to the game's setting, which differs from that of a conventional laboratory game played between anonymous strangers. However, while the minimum effort levels were stable throughout, they were not identical across different types of groups. Like in the public goods game, the most efficient outcomes were achieved by those with highest cohesion development, and the least efficient outcomes by those with the lowest.

Figure 4: Efficiency in the weak-link game



Note: In the low partition (19 groups), change in group cohesion < 1; in the medium partition (74 groups), change in group cohesion is between 1 and 1.25; in the high partition (91 groups), change in group cohesion > 1.25.

These patterns support the importance of group cohesion as a determinant of efficient group interaction. However, since the analysis above demonstrated few treatment differences in the growth of group cohesion, we might expect similarly few treatment differences in the game outcomes. Figures 5 and 6 plot the average earnings over time in the public goods and weak-link games for each treatment. In most cases the outcomes look similar across treatments, although efficiency appears to be relatively high in the *Between-group-inequality* treatment, especially for the weak-link game.

To formally address our hypotheses, we run regressions with the game outcomes as the dependent variables. These are presented in Table 7. Each individual is treated as one observation, and standard errors are clustered at the group level. To test Hypothesis 4, Models (1) and (2) include only the variable capturing group cohesion change, as well as demographic

Figure 5: Contributions in the public goods game by treatment

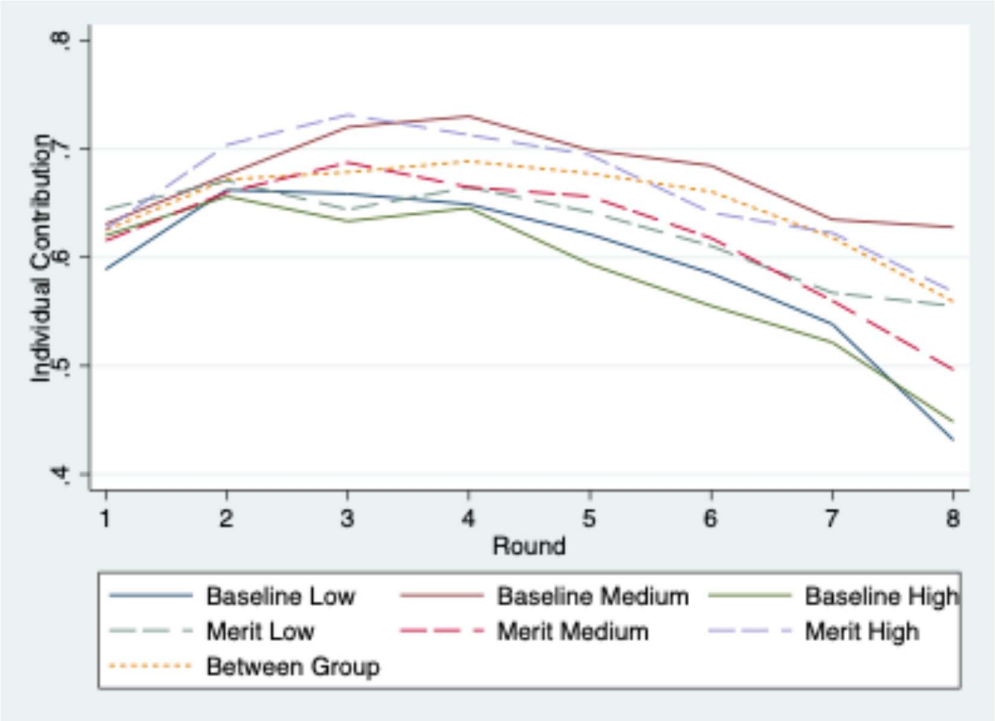


Figure 6: Earnings in the weak-link game by treatment

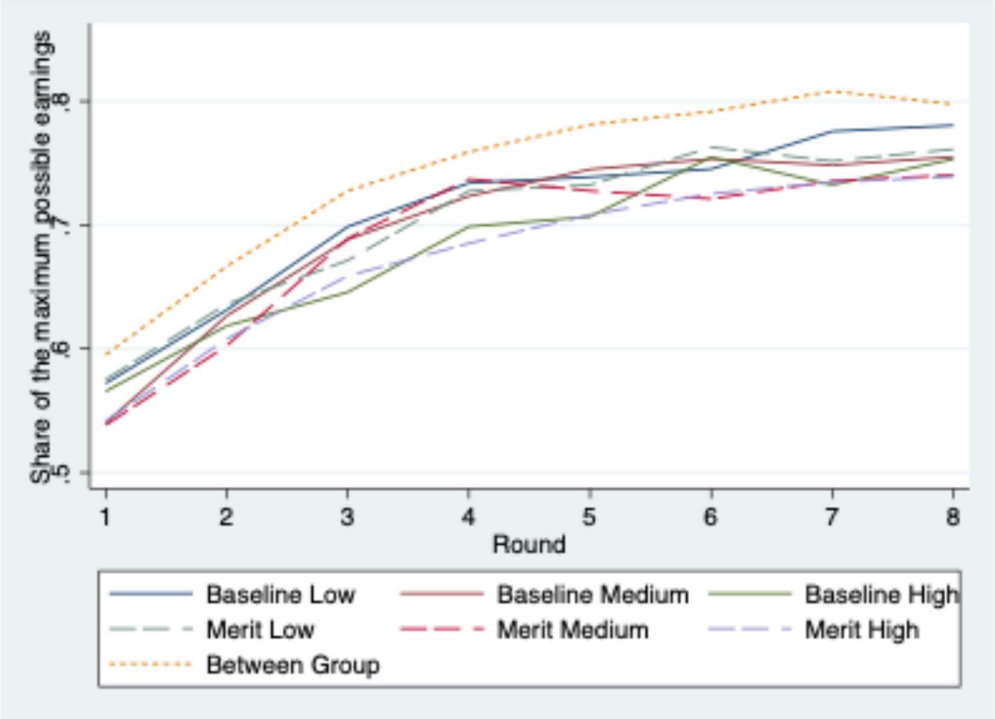


Table 7: Treatment effects and mediating effects of group cohesion on game outcomes

	Game Outcome		Game Outcome		Game Outcome	
	(1)	(2)	(3)	(4)	(5)	(6)
	Public Goods	Weak Link	Public Goods	Weak Link	Public Goods	Weak Link
The change in group cohesion	3.048*	4.598			2.753*	3.958
	(2.49)	(1.15)			(2.07)	(0.94)
Treatment:						
Baseline Low			ref.	ref.	ref.	ref.
			(.)	(.)	(.)	(.)
Baseline Medium			3.047	-2.733	1.746	-4.188
			(1.11)	(-0.36)	(0.63)	(-0.55)
Baseline High			-0.670	-5.971	-0.994	-6.433
			(-0.26)	(-0.69)	(-0.40)	(-0.75)
Between Group			2.376	6.940	1.051	5.031
			(0.92)	(0.83)	(0.39)	(0.58)
Merit Low			1.452	-0.735	1.109	-1.233
			(0.55)	(-0.09)	(0.42)	(-0.15)
Merit Medium			1.070	-4.514	0.978	-4.649
			(0.42)	(-0.59)	(0.38)	(-0.61)
Merit High			2.698	-6.611	1.839	-7.845
			(1.11)	(-0.84)	(0.75)	(-0.99)
Control:						
Age	0.469	1.702*	0.549*	1.968**	0.485	1.865**
	(1.70)	(2.55)	(2.00)	(2.85)	(1.75)	(2.80)
Gender	-0.900	-3.720	-0.772	-3.721	-0.797	-3.772
	(-0.81)	(-1.31)	(-0.71)	(-1.34)	(-0.74)	(-1.35)
Constant	74.25***	93.84***	75.51***	96.71***	73.46***	94.02***
	(12.42)	(5.94)	(12.19)	(6.11)	(11.95)	(5.74)
Observations	722	734	722	734	722	734

Notes: OLS regressions are presented. The game outcomes are contributions in the public goods game and earnings in the weak-link game. t statistics in parentheses. Standard errors are clustered at the group level. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

controls. Model (1) finds that groups with stronger development of group cohesion achieved significantly higher public goods contributions. Model (2) also finds a positive relationship between group cohesion development and earnings in the weak-link game, in line with the evidence from Figure 4, but this is not significant. Therefore, part (a) of Hypothesis 4 is supported, while there is not sufficient evidence to support part (b).

Models (3) and (4) regress the game outcomes against the treatment dummies, alongside demographic controls. With the *Low-inequality Baseline* as the reference category, none of the treatment dummies are significant, either for public goods contributions or weak-link game earnings. We furthermore run linear restriction tests for all treatment pairs that are the subject of our hypotheses, which find no significant differences for either outcome variable. Therefore, Hypothesis 5 – that higher within-group inequality would result in less efficient outcomes – is not supported. Hypothesis 6 – that higher between-group inequality would result in more efficient outcomes – is also not supported, despite the directional effects in the expected direction that we saw in Figures 5 and 6. Hypothesis 7 is also not supported, as there are no significant outcome differences between *Baseline* and corresponding *Merit-based* treatments.

Note that models (3) and (4) do not control for change in group cohesion and therefore identify the total effects of the treatment variables inclusive of the influence they exert on group cohesion. Models (5) and (6), which do control for the change in group cohesion variable, identify the partial treatment effects independent of a mediating effect via group cohesion. Not surprisingly, in these models the treatment dummies – and linear restriction tests – remain insignificant.[14]

Robustness Checks

First, we assess the similarity of results across universities. Table E1 replicates the analysis of Table 6, separated between the Ningbo and Wuhan sessions. This shows that, at both universities, between-group inequality resulted in significantly stronger development of group cohesion, while other treatment effects on this outcome are generally insignificant. Table E2 and E3 replicate Table 7 for each sample. The results are qualitatively similar for the most part, with an absence of treatment effects on the public goods and weak-link game outcomes. One

[14] Re-running these analyses with our alternative measures of group cohesion produces qualitatively similar results.

exception is that the effects of the *Between-group-inequality* treatment on weak-link game earnings go in opposite directions at the two universities, seemingly driven by an anomalous (insignificant) negative relationship between development of group cohesion and game efficiency in Ningbo.

A key difference between our study and traditional lab experiments is that, with anonymity removed, the payoffs subjects leave the lab with may not be the true final payoffs. In principle, those subjects who benefit from the unequal pay rates could promise to compensate their impoverished team-mates by sharing part of their earnings after the experiment has ended. While this is a perfectly realistic feature of social relationships under inequality, such retrospective redistribution is not observable to us and we would like to gain some understanding of whether such promises took place and, if so, whether they play a part in explaining the absence of harmful effects of within-group inequalities.

We included a question, at the very end of the experiment, which asked subjects: "During the experiment, did your group at any point discuss the possibility of group member who earned relatively high amounts of money from the slider task sharing some of their earnings with those group members who earned relatively low amounts of money from the slider task?". Only 6.25% of subjects answered the question in the affirmative, suggesting that the impact of such redistribution was likely small. Furthermore, Table E4 shows no dramatic differences across treatments, with the percentage of affirmative responses ranging from 3.6% to 10.6%. Finally, in Tables E5 and E6, we re-run our main analyses from Tables 6 and 7, excluding those subjects who answered the question positively, and obtaining qualitatively unchanged results.

4. Conclusion

This paper has examined whether economic inequality influences the efficiency of group interaction by shaping the development of group cohesion. Using a novel experimental design that combined prolonged face-to-face interaction, experimentally manipulated inequalities in earning opportunities, direct measurement of group cohesion, and incentivised measures of cooperative and coordinated behaviour, we investigated inequalities both within and between groups.

Our results paint a nuanced picture. We find that groups whose members develop stronger cohesion subsequently interact more efficiently, particularly in cooperative settings. This

broadly supports and extends the findings of Gächter et al. (2025). In contrast to their study, we find the relationship between group cohesion development and efficient coordination game outcomes does not reach conventional levels of statistical significance, but it is directionally present. Overall, our results support the economic importance of group cohesion.

However, we find little evidence that inequalities within groups inhibit the formation of these social ties. Even when differences in earnings opportunities were extremely large, groups generally formed close interpersonal relationships, and neither the magnitude of within-group inequality nor whether it was determined by luck or merit had substantial effects on cohesion or on subsequent efficiency. These findings suggest that interpersonal relationships may be considerably more resilient to inequality than we initially expected.

Furthermore, we find that inequality between groups strengthens the development of cohesion within groups. This provides evidence for a mechanism that has received little attention in the economics literature. While previous studies have primarily emphasised the harmful consequences of between-group inequality – including increased conflict, instability and reduced cooperation across groups (Brañas-Garza et al., 2021; Fallucchi et al., 2021; Eckel et al., 2022; Baier et al., 2024; Barr et al., 2024; Levi and Ramalingam, 2025) – our findings suggest that the same inequalities can simultaneously reinforce solidarity among members of individual groups. In settings where productive activity occurs largely within relatively homogeneous teams, this mechanism may partly offset some of the channels through which inequality is argued to reduce economic performance. As a caveat, we note that while our experiment found a strongly significant positive effect of between-group inequality on the development of group cohesion, its effects on game efficiency did not reach conventional levels of statistical significance. Nevertheless, the estimated treatment effects on efficiency were consistently in the expected direction. We note that the relationship between group cohesion and game outcomes is itself somewhat noisy (and not significant for the weak-link game), potentially reducing statistical power to detect a treatment effect mediated by this channel. Importantly, however, our results do provide clear evidence that between-group inequality influences the proposed mediating variable, namely the development of group cohesion.

More broadly, our findings present a challenge to arguments that inequality necessarily harms efficiency through its effects on social relationships. We do not argue that inequality is economically or socially benign. Our design was intentionally focused on one particular mechanism – the development of group cohesion – and inequality may well affect efficiency

through many other channels that were beyond the scope of this study. Nevertheless, our results indicate that this particular channel appears surprisingly robust against harmful consequences from inequality. At the same time, they emphasise that understanding the consequences of inequality requires distinguishing carefully between different forms of inequality and the different social mechanisms through which they may operate.

Like any study, ours also has limitations. Our participants were university students, interacting over the course of a half-day programme in an experimental environment designed to facilitate social interaction. Future research should examine whether similar patterns emerge in naturally occurring workplaces, organisations and communities, where relationships develop over much longer periods and where inequalities may coexist with additional institutional and social factors.

Our research was conducted in China. The consistency of the results across two socioeconomically and culturally distinct university populations provides some reassurance regarding external validity, but how groups react to inequality may be subject to cultural factors which differ across populations. This is not to downplay the importance of studying our research questions in China, the world's second largest economy and second most populous country. Nevertheless, future studies examining whether our findings obtained in a particular non-WEIRD setting can be replicated in WEIRD countries (Henrich et al., 2010), or elsewhere, would also be of value.

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Appendix

Appendix A: Experimental Instructions – English translation

Below, we present the instructions – translated into English – from the *Baseline low inequality* treatment. Sections of text that differ in other treatments are highlighted for the reader's attention (this highlighting was not present in the instructions subjects received). Footnotes report the treatment variations in the preceding highlighted text.

The instructions for each activity were distributed only after the previous activity (or questionnaire) had been completed.

Introduction

Welcome to this experiment. This is an experiment about decision-making. There will be several stages to the experiment. First, there will be a series of activities that we refer to as Phase 1. This part of the experiment will take around 3 hours. This will include a break in which you will be provided a meal. Afterwards, we will move onto Phase 2, in which you will do a computer-based activity.

At the end of the experiment, you will receive payment. This payment will depend upon the outcome of several activities:

- 3 activities in Phase 1
- The computer-based activity in Phase 2
- In addition, everyone will earn 20 RMB as a base payment. This will be added to your payments from the other activities

You will discover the amounts you have earned from each activity as we go through the experiment. However, you will not actually receive the money until the experiment finishes. In total, the experiment will take around 4 hours to complete. Please note that once the experiment begins, you will not be able to withdraw.

In this experiment, you have been placed into a team consisting of the 4 people sitting at your table now. You will stay in the same team for all activities in the experiment. In some tasks in Phase 1 of the experiment, there will be different roles that different members of the team are assigned into. In Phase 2 of the experiment, all team members will be given IDENTICAL roles. Therefore, note that the roles you are given in Phase 1, and the outcome of the activities in Phase 1, will NOT have any influence on your assigned role or the nature of your assigned activity in Phase 2.

If at any point you need anything explained to you, please raise your hand and the experimenter will help you.

First, we ask you to complete a questionnaire. This questionnaire is confidential so don't confer with your team members or look at their answers.

……..

Activity 1

In this activity, you are given a task to perform, from which you can earn money.

For all team members, the task is the same. You are provided with a tablet. On the tablet, you must complete five rounds of the task in order to score 'points'. Each round will be identical and will last 120 seconds. The round will consist of a screen with 48 sliders. Each slider is initially positioned at 0 and can be moved as far as 100. Each slider has a number to its right showing its current position. You can use your finger (or the mouse) in any way you like to move each slider. You can readjust the position of each slider as many times as you wish. Your points in each round are given by the number of sliders positioned at exactly 50 at the end of the 120 seconds. Your total points from this task will be given by the sum of your points earned from each round.

Although the task is the same for all team members, the rate of pay is not the same. According to your randomly assigned seat, you will be given one of four roles. If your seat is 1, 5, 9, or 13, you will be assigned to "Role 4". If your seat is 1, 5, 9, or 13, you will be assigned to "Role 3". If your seat is 2, 6, 10, or 14, you will be assigned to "Role 2". If your seat is 4, 8, 12, or 15, you will be assigned to "Role 1". Your role will determine your pay rate, as follows:[15]

[15] In the Merit-based inequality treatments, this text read: "Although the task is the same for all team members, the rate of pay is not the same. Before the activity begins, all members of your team will play one "qualifying round" of this slider task. In this qualifying round you cannot earn money, but you will be ranked according to your performance in it. The team member with the most points in this round will be assigned to "Role 1"; the team member with the second most point will be assigned to "Role 2"; the team member with the third most points will be assigned to "Role 3"; and the team member with the least points will be assigned to "Role 4". If two or more team members score the same number of points, their relative rank will be determined randomly. Your role will then determine your pay rate in all future rounds of this activity, as follows:"

In the Between-group inequality treatment, it read: "Although the task is the same for all participants, the rate of pay is not the same. Before the task begins, the computer software will randomly assign the four teams in the room to one of four different pay levels. One of the four teams will be assigned to Pay Level 1, another to Pay Level 2, another to Pay Level 3, and another to Pay Level 4. Within each team,

- If you are assigned to “Role 1”, your earnings from the task are 0.45 RMB per point
- If you are assigned to “Role 2”, your earnings from the task are 0.44 RMB per point
- If you are assigned to “Role 3”, your earnings from the task are 0.43 RMB per point
- If you are assigned to “Role 4”, your earnings from the task are 0.43 RMB per point[16]

Before this activity begins[17], you will first get the chance to have 3 practice rounds at the slider task. These rounds are purely for the purpose of helping you get used to the task. You cannot earn money from these rounds, and your performance in these rounds has no effect on the rest of the experiment.

When everyone is ready, we will announce the beginning of the first practice round. Are there any questions?

---

the computer software will then randomly assign each member to either “Role 1”, “Role 2”, “Role 3” or “Role 4”. Your team’s Pay Level and your individual role will determine your pay rate, as follows:”

[16] The pay rates varied across treatments. In the Between-group inequality treatment, this section of text separately listed the pay rate each role would obtain for each of the four group Pay Levels they could be assigned.

[17] In the Merit-based inequality treatments, this text read “Before the qualifying round”.

Activity 2

The purpose of the next activity is for you to get to know more about your team members.

Below is a list of questions. Your team should consider these questions and discuss them as a group. Your conversation can take any form that you choose. The aim is just for you and your team members to find out more about each other. You have 10 minutes for this activity.

Discuss the following about yourself and your team members

1. **What is your major? Which year of study are you in?**
2. **Where did you grow up?**
3. **Why did you sign up to participate in this study?**
4. **When you are extremely anxious, what do you like to do to relax?**
5. **If you could meet any living person for dinner, who would you pick and why?**
6. **If you had 1 billion RMB, how would you spend it?**
7. **Would you rather be invisible or be able to fly?**
8. **What is your dream job?**
9. **If you had to delete all but three apps from your smartphone, which ones would you keep?**

……..

Activity 3

In this activity you are asked to come up with an idea for a new smartphone app. You need to work together with your team members to design the app. You have 10 minutes to come up with the idea. Afterwards, each team needs to briefly present the idea for their app to the other teams in the room (aim to speak for about 2 minutes).

Some things to think about:

- What is the purpose of your app?
- How would it work?
- Why do you think this would be useful?
- What would be the target demographic for the app?
- What challenges would there be in making this app a success?

……..

Activity 4

In this activity, you will again perform the same task as in Activity 1. Again, you can earn money from this task.

The rules of this task are exactly the same as in Activity 1. There will be 5 rounds consisting of 120 seconds each. Your points in each round will be given by the number of sliders on exactly 50 at the end of the round. Your total points will be given by the sum of your points from each round.

As before, your rate of pay depends upon your role. You will keep the same role as in Activity 1.[18] To remind you:

- If you are assigned to "Role 1", your earnings from the task are 0.45 RMB per point
- If you are assigned to "Role 2", your earnings from the task are 0.44 RMB per point
- If you are assigned to "Role 3", your earnings from the task are 0.43 RMB per point
- If you are assigned to "Role 4", your earnings from the task are 0.43 RMB per point[19]

[18] In the Between-group inequality treatment, this text reads: "As before, your rate of pay depends upon your team's pay level and your individual role. You will keep the same team pay level and individual role as in Activity 1."

[19] These instructions vary across treatments, as for Activity 1.

Activity 5

In this activity, you will again perform the same task as in Activity 1 and Activity 4. Again, you can earn money from this task.

The rules of this task are exactly the same as in Activity 1 and Activity 4. There will be 5 rounds consisting of 120 seconds each. Your points in each round will be given by the number of sliders on exactly 50 at the end of the round. Your total points will be given by the sum of your points from each round.

As before, your rate of pay depends upon your role. You will keep the same role as in Activity 1 and Activity 4.[20] To remind you:

- If you are assigned to "Role 1", your earnings from the task are 0.45 RMB per point
- If you are assigned to "Role 2", your earnings from the task are 0.44 RMB per point
- If you are assigned to "Role 3", your earnings from the task are 0.43 RMB per point
- If you are assigned to "Role 4", your earnings from the task are 0.43 RMB per point[21]

[20] In the Between-group inequality treatment, this text reads: "As before, your rate of pay depends upon your team's pay level and your individual role. You will keep the same team pay level and individual role as in Activity 1."

[21] These instructions vary across treatments, as for Activity 1.

Activity 6

This activity is a competition between teams. You will be given a series of quiz questions. The experimenter will read out each question. You should discuss the question with your team members and write down your answer on the sheet of paper provided. You are NOT allowed to use your phones during this task to look up the answers to these questions. The team who achieves the most correct answers will be the winner.

Please also write down a name for your team.

Your team name:

Round 1 answers	Round 2 answers
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.
10.	10.

……

Phase 2

In this stage of the experiment, you will participate in three activities. At the end, one of these three activities will be selected at random and the total amount of RMB you have earned in that activity will be paid to you for this stage of the experiment. This will be added to your earnings from the earlier stages of the experiment to determine the final amount you will earn today.

As you will not know which of these three activities will count towards your earnings until the end of the experiment, please make your decisions in each activity carefully.

During the activities, you must not speak or attempt to communicate with your team members.

Activity 7[22]

In this activity you are asked to make three decisions.

In each of these decisions you are initially assigned 160 RMBs. You need to decide how much of this amount you will keep for yourself, and how much of this amount you will donate to a charity specified in each decision below.

At the end of this activity, one of the three decisions below will be selected at random, and the amount you chose to keep for yourself will be your earnings of this activity. The other two decisions will not count towards your earnings. If at the end of the experiment this activity is selected for determining your earnings from Phase 2, the donation decision of this activity will be executed. The amount you donate in this decision will be given to the charity as an anonymous donation, and the amount you chose to keep for yourself will be your earnings from this stage of the experiment.

As any of these decisions might count towards your earnings, please make each decision carefully.

Decision 1

You are assigned 160 RMBs. Please indicate how much of this amount you will keep for yourself. The money you do not keep will be donated to a local charity in Ningbo[23].

[22] Note that the order of Activities 7, 8 and 9 was randomised across sessions.

[23] This text was used for sessions run in Ningbo. For sessions run in Wuhan, subjects were informed that the local charity was in Wuhan.

Decision 2

You are assigned 160 RMBs. Please indicate how much of this amount you will keep for yourself. The money you do not keep will be donated to a national charity in China.

Decision 3

You are assigned 160 RMBs. Please indicate how much of this amount you will keep for yourself. The money you do not keep will be donated to an international charity.

Activity 8

In this activity you are asked to make a series of decisions. There will be eight rounds. In every round, you are given 8 RMB, and you have to decide how much of it you want to contribute to a project and how much of it to keep for yourself. Each of the other three members of your group has also been given 8 RMBs and has to make the same decision, i.e. they have to decide how many of their 8 RMBs they want to contribute to the project and how many of them to keep for themselves. Every RMB that you do not contribute to the project automatically belongs to you and earns you one RMB. For the RMBs contributed to the project the following happens: the total contributions to the project will be calculated by adding up all individual contributions of the group members. The total contribution will then be multiplied by 1.6 and this amount will be divided equally among all four members of the group. For example, if a total of 2 RMBs is contributed to the project, the project's value increases to 3.2 RMB. This amount is divided equally among all four members of the group. Thus, every group member receives 0.8 RMBs.

Your total earnings for each round therefore consists of two parts:

(1) The RMBs which you have kept for yourself ("Earnings from retained RMBs"),

(2) The "Earnings from the project".

The calculation of the "Earnings from the project" is equal to 0.4 *times* the total contributions from all group members to the project (the number 0.4 is obtained by dividing the 1.6 multiplication factor by 4, which is the total number of members in your group). The earnings from the project of every other group member is calculated in the same way, i.e., each group member receives the same earnings from the project.

Your earnings for each round is therefore:

(8 RMBs – Number of RMBs contributed to the project by you) + 0.4*(Sum of all RMB contributed to the project by all members of your group)

Note that you will be paid the sum of your earnings from all eight rounds. After each round, you will receive information about the total contributions to the project by all group members, and your earnings, from the previous round.

Activity 9

In this activity you are asked to make a series of decisions. There will be eight rounds. In every round, each member of your group including you will choose a value of X. The values of X you may choose are 1, 2, 3, 4, 5. The value you choose for X and the smallest value of X chosen by any participant in your group will determine the earnings you receive for that round. The table below (EARNINGS TABLE) shows the potential earnings that you may receive. Please look at this table now. Note that the EARNINGS TABLE is the same for every participant in the experiment. Earnings are in RMB. Only your earnings are shown in the EARNINGS TABLE. The earnings in each round may be found by looking across from the value you choose on the left-hand side of the table and down from the smallest value of X chosen by a participant in your group. For example, suppose the other members of your group choose 4, 5 and 4 respectively while you choose 3. Thus, the smallest value chosen is 3 and you earn 20 RMB for that round. As a second example, if you choose 2 and the smallest value chosen is 1, you earn 12 RMB for that round. Note that you will be paid the sum of your earnings from all eight rounds. After each round, you will receive information about the smallest value of X chosen in your group, and your earnings, from the previous round.

EARNINGS TABLE (per round)

		The smallest value of X chosen				
		1	**2**	**3**	**4**	**5**
	1	16				
	2	12	18			
Your	**3**	8	14	20		
choice	**4**	4	10	16	22	
	5	0	6	12	18	24

Appendix B: Experimental Instructions – Chinese version

Below are the original Chinese instructions from the *Baseline low inequality* sessions.

导语：

欢迎参加本实验。这个是一个关于决策制定的实验，将分为几个不同的阶段进行。首先将会有一系列我们称之为第一阶段的活动。这部分实验大约需要 3 个小时，其中包含一段休息时间，期间我们将会为你提供餐食。之后我们将进入第二阶段，进行一些基于计算机的活动。

你在实验结束后会收到一定的报酬。具体金额将取决于几项活动的结果：

- 1.第一阶段中的三场活动；
- 2.第二阶段中基于计算机的活动；
- 3.此外，每个人都将获得 20 元人民币的基础报酬。这部分将被加入你从其他活动所获得的报酬。

随着实验的进行，你会得知你参与每一项活动所获得的报酬。但是，只有实验完成，你才会真正收到报酬。完成实验总共需要大约 4 个小时。请注意**实验开始，将无法退出**。

在本实验中，你将与另外三名同桌成员组成四人团队，本分组将贯穿这次实验的所有活动。在第一阶段中进行的部分任务中，你和其他团队成员将分配到不同的角色。而在第二阶段，所有成员都会被赋予**完全相同的**角色。因此，请注意，你在第一阶段被分配的角色及其活动结果不会对之后第二阶段的角色或者任务分配产生任何影响。

在任何时候如有疑问，请举手，实验人员将会为你提供帮助。

首先，请完成一份问卷。由于此问卷是保密的，所以请勿和团队成员讨论或查看他人的答案。

……

活动 1

在这场活动中，你将会被分配一项任务并可以从中获得报酬。

所有团队成员的任务都相同。我们会为你提供一台平板电脑。你必须在平板电脑上完成五轮任务。每一轮任务内容和时长都完全相同。每轮任务开始时，屏幕会显示 48 个滑动条。每个滑动条上都有一个滑块，右边的数字则显示滑块当前位置。

你可以用手指(鼠标)随意调整每个滑块的位置。滑块最左可移动至 0，最右可移动至 100。你有 120 秒的时间来将所有滑块移动至 50，计时结束后，电脑将按位置处于 50 的滑块数量来计算每轮积分。你在这场活动中的总分是五轮任务中获得的积分的总和。

虽然所有团队成员的任务都是一样的，但工资率却不一样。任务开始前，根据你被随机分配的座位，你将被分配四个角色中的一个。如果你的座位号是 1、5、9 或 13，你将被分配到“角色 4”。 如果你的座位号是 2、6、10 或 14，你将被分配到“角色 3”。 如果你的座位号是 3、7、11 或 15，你将被分配到“角色 2”。 如果你的座位号是 4、8、12 或 16，你将被分配到“角色 1”。你的角色将决定你的工资率，如下所示:

- 如果你被分配到“角色 1”，你从任务中获得的收入是每积分 0.45 元人民币
- 如果你被分配到“角色 2”，你从任务中获得的收入是每积分 0.44 元人民币
- 如果你被分配到“角色 3”，你从任务中获得的收入是每积分 0.43 元人民币
- 如果你被分配到“角色 4”，你从任务中获得的收入是每积分 0.43 元人民币

在活动开始之前，你将有机会在滑块任务中进行 3 轮练习。 这些纯粹是为了帮助你熟悉任务。 你不能从这几轮练习中赚取报酬，而你在期间的表现对实验的其余部分没有影响。

当每个人都准备好后，我们将宣布第一轮练习开始。

大家是否有问题?

活动 2

下一场活动的目的是让你更多地了解你的团队成员。

下面是一张问题清单。每个团队应该仔细考虑并以小组形式互相交流，讨论这些问题。交流方式任你选择，目的仅在于让你和你的团队成员更好地了解彼此。本活动限时十分钟。

讨论以下关于你本人和其他团队成员的问题：

1.你的专业是什么？读第几年了？

2.你在哪里长大？

3.你为什么报名参加这项研究？

4.当你极度焦虑时，你喜欢做什么来放松？

5.如果你可以和任何在世的人共进晚餐，你会选谁？为什么？

6.如果你有 10 亿人民币，你会怎么花？

7.你是想隐身还是能够飞行？

8.你梦想的工作是什么？

9.如果只能保留三个你手机上的 APP 并删除所有其他 APP，你会选择保留哪三个？

……

活动 3

在这场活动中，你需要提出一个设计智能手机 APP 的新想法。你需要和你的团队成员一起设计这款 APP，限时 10 分钟。之后每个团队需要向房间里其他团队简要介绍这款 app 的想法(汇报时长约 2 分钟)。

一些需要考虑的事:

- 你这款应用程序的目的是什么?
- 它如何运作?
- 你为什么认为它会有用?
- 这个 APP 的目标人群是谁?
- 要使这款 APP 获得成功会遇到哪些挑战?

.....

活动 4

在这场活动中，你将再次执行与活动 1 中相同的任务。同样，你可以从本任务中赚取报酬。

规则与活动 1 完全相同。共有 5 轮，每轮 120 秒。120 秒后本轮结束，按位置处于 50 的滑块数量计算每轮积分。你在这场活动中的总分是每轮中获得的积分的总和。

与之前相同，你的工资率取决于你的角色。你的角色也与活动 1 中的相同。

提示：

- 如果你被分配到“角色 1”，你从任务中获得的收入是每积分 0.45 元人民币
- 如果你被分配到“角色 2”，你从任务中获得的收入是每积分 0.44 元人民币
- 如果你被分配到“角色 3”，你从任务中获得的收入是每积分 0.43 元人民币
- 如果你被分配到“角色 4”，你从任务中获得的收入是每积分 0.43 元人民币

.....

活动 5

在这场活动中，你将再次执行与活动 1 及活动 4 相同的任务。同样，你可以从本任务中赚取报酬。

任务规则与活动 1 和活动 4 的规则完全相同。共有 5 轮，每轮 120 秒。120 秒后本轮结束，按位置处于 50 的滑块数量计算每轮积分。你在这场活动中的总分是每轮中获得的积分的总和。

与之前相同，你的工资率取决于由你的角色。你的角色依旧保持与活动 1，4 中的一致。

提示：

•如果你被分配到“角色 1”，你从任务中获得的收入是每积分 0.45 元人民币

•如果你被分配到“角色 2”，你从任务中获得的收入是每积分 0.44 元人民币

•如果你被分配到“角色 3”，你从任务中获得的收入是每积分 0.43 元人民币

•如果你被分配到“角色 4”，你从任务中获得的收入是每积分 0.43 元人民币

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活动 6

这场活动是团队之间的竞赛。这里有一系列知识问答题。实验人员将读出每一个问题。你应该与你的团队成员讨论此问题，并将答案写在提供的纸上。这项任务过程中你被**禁止**使用手机查找答案。完成最多正确答案的小组将获胜。

请写下你的队名：

你的队名：

第一轮的答案	**第二轮的答案**
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.
10.	10.

……

第二阶段

在实验的这一阶段，你将参加三场活动。在最后，这三场活动仅有一场会被随机抽中并支付报酬，你参与该场活动获得的收入即为你在本阶段的报酬。加上你在先前阶段的收入，这就是你今天赚取的最终收益。

由于你在实验结束前不会知道这三项活动中哪一项将计入你的收入，请在每场活动中都仔细地做决定。

在这些活动中，你不得与团队成员交谈或尝试交流。

活动 7

在这场活动中，你需要做出三个决定。

在每一次做出决定前，你会分到 160 元。你需要决定其中有多少留给自己，以及有多少将捐赠给在下列每个决定中所指定的慈善机构。

在本活动结束时，以下三个决定中的一个将被随机选中，而你在这个决定中为自己保留的金额则是你这场活动的收入，其他两个捐赠决定将不会计入你的收入。如果在实验结束时，本活动被选中来计算你在第二阶段的收益，届时这场活动的捐赠决定将被执行。你在此项决定中捐赠的金额将以匿名捐赠的形式捐赠给该慈善机构，而你选择为自己保留的金额则是你在此实验阶段的收入。

所以这些捐赠决定可能会影响到你的收入，请仔细做出每一个决定。

决定 1

你分到 160 元人民币。请说明你将为自己留下的金额。你未保留的钱将捐给宁波本地的一家慈善机构。

决定 2

你分到 160 元人民币。请说明你将为自己留下的金额。你未保留的钱将捐给中国的一家国家级慈善机构。

决定 3

你分到 160 元人民币。请说明你将为自己留下的金额。你未保留的钱将捐给一家国际慈善机构。

活动 8

在这场活动中，你需要做出一系列决定。共 8 个回合。你在每一轮将得到 8 元人民币，之后你必须决定你想为一个项目投入多少和为自己留下多少。同一团队的其他三名成员每人也得到 8 元并需要做相同的决定，即必须决定这 8 元人民币中有多少投入项目，多少留给自己。未投入到项目中的钱自动归你所有。对于投入项目的钱，会发生以下情况：对项目的总投入为所有团队成员每个人投入的总和。之后，将总投入将乘以 1.6，再平分给同组的四个成员。例如，如果项目投入总额为 2 元，则项目价值增至 3.2 元，再平分给该小组的四个成员。即，每个小组成员得 0.8 元人民币。

因此，你每轮的总收入包括两部分:

(1)为自己留存的钱(“留存人民币所得”)；

(2)“项目收入”。

“项目收入”的计算方法为 0.4 *乘以*小组成员对项目的总投入（0.4 这个数字是用 1.6 除以 4 得到的，因为小组有 4 个人)。其他每一位小组成员的项目收入的计算方法相同，每位小组成员从项目中获得的收入是相同的。

因此，每一轮的收入为:
(8 元-你投入到项目的金额)+ 0.4*(团队所有成员对项目投入金额的和)

请注意，你的收入是所有八轮的收入的总和。每轮结束后，你将得知团队所有成员对项目投入的总金额以及你上一轮的收入。

活动 9

在这场活动中，你需要做出一系列决定。一共 8 个回合。在每一轮中，包括你在内的所有小组成员将各自选择一个值 X，X 的取值范围是 1、2、3、4、5。你选择的 X 值和所在小组所有成员选择的 X 最小值将决定你在这一轮中获得的收入。下表(收入表)显示了你可能获得的收入。现在请仔细看这张表。注意，此收入表对于实验中的每个参与者都是相同的。收入以人民币计算。此收入表中只显示你的收入。表格纵轴为你选择的 X 值，横轴为你所在小组所选的最小 X 值，二者相交处即为你每一轮的收入。例如，假设同组的其他成员分别选择 4、5 和 4，而你选择 3。这样，所选 X 的最小值是 3，这一轮你将获得 20 元人民币。再举个例子，如果你选择了 2，并且本组所选最小值是 1，那么此轮你将获得 12 元人民币。请注意，你将获得所有八轮收入的总和。每轮结束后，你将得知你的团队中选择的 X 的最小值以及你上一轮的收入。

收入表(每轮)

		所选 X 的最小值				
		1	**2**	**3**	**4**	**5**
你的选择	**1**	16				
	2	12	18			
	3	8	14	20		
	4	4	10	16	22	
	5	0	6	12	18	24

Appendix C: Alternative measures of group cohesion and full Wilcoxon rank-sum test results

Table C1: The levels of group cohesion before and after the slider task by treatment: alternative measures of group cohesion

	Treatment							
	Baseline Low	Baseline Medium	Baseline High	Between Group	Merit Low	Merit Medium	Merit High	Total
Alternative 1: The average minimum rating each player receives								
Pre-task	1.087	1.070	1.019	1.027	1.135	1.010	1.130	1.068
	(0.211)	(0.184)	(0.098)	(0.079)	(0.476)	(0.049)	(0.263)	(0.237)
Post-task	2.000	2.220	2.163	2.375	2.240	2.077	2.472	2.224
	(0.524)	(0.659)	(0.552)	(0.661)	(0.594)	(0.423)	(0.606)	(0.591)
Change	0.913	1.150	1.144	1.348	1.106	1.067	1.343	1.156
	(0.463)	(0.591)	(0.539)	(0.654)	(0.511)	(0.422)	(0.551)	(0.549)
Alternative 2: The minimum of all 12 ratings								
Pre-task	1.000	1.000	1.000	1.000	1.077	1.000	1.037	1.016
	(0.000)	(0.000)	(0.000)	(0.000)	(0.272)	(0.000)	(0.192)	(0.127)
Post-task	1.769	1.880	1.885	2.071	2.077	1.846	2.222	1.967
	(0.587)	(0.666)	(0.516)	(0.663)	(0.628)	(0.464)	(0.641)	(0.609)
Change	0.769	0.880	0.885	1.071	1.000	0.846	1.185	0.951
	(0.587)	(0.666)	(0.516)	(0.663)	(0.566)	(0.464)	(0.622)	(0.594)
Alternative 3: The average of all 12 ratings								
Pre-task	1.670	1.687	1.692	1.622	1.676	1.628	1.759	1.676
	(0.547)	(0.675)	(0.596)	(0.551)	(0.670)	(0.450)	(0.642)	(0.586)
Post-task	2.933	3.250	3.119	3.336	3.029	3.000	3.383	3.153
	(0.642)	(0.763)	(0.708)	(0.615)	(0.612)	(0.552)	(0.727)	(0.672)
Change	1.263	1.563	1.426	1.714	1.353	1.372	1.623	1.476
	(0.465)	(0.712)	(0.655)	(0.512)	(0.431)	(0.489)	(0.473)	(0.554)

Note: Standard deviations in parentheses

Table C2: Significance of treatment differences for change in group cohesion – Wilcoxon rank-sum tests

	Main Version	Alternative 1	Alternative 2	Alternative 3
Hypothesis 1				
Low-inequality Baseline vs Medium-inequality Baseline	0.022*	0.135	0.622	0.085
Low-inequality Baseline vs High-inequality Baseline	0.184	0.047*	0.424	0.249
Medium-inequality Baseline vs High-inequality Baseline	0.365	0.811	0.768	0.546
Low-inequality Merit-based vs Medium-inequality Merit-based	0.512	0.985	0.300	0.687
Low-inequality Merit-based vs High-inequality Merit-based	0.365	0.189	0.382	0.048*
Medium-inequality Merit-based vs High-inequality Merit-based	0.148	0.183	0.034*	0.130
Hypothesis 2				
Low-inequality Baseline vs Between-group-inequality	0.000***	0.003**	0.097	0.001***
Hypothesis 3				
Low-inequality Baseline vs Low-inequality Merit-based	0.049*	0.099	0.151	0.203
Medium-inequality Baseline vs Medium-inequality Merit-based	0.256	0.766	0.933	0.418
High-inequality Baseline vs High-inequality Merit-based	0.318	0.570	0.060	0.200

Note: p-values are presented. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Four Measurements for Group Cohesion:

*Main version: the average of min ratings each player gives to group members

*Alternative 1: the average minimum rating each player receives from group members

*Alternative 2: the minimum of all 12 ratings

*Alternative 3: the average of all 12 ratings

Appendix D: Dyadic relationships

Table D1: The changes in closeness for each dyadic relationship

	The change in reported closeness toward			
Low-inequality Baseline				
– From:	Role 1	Role 2	Role 3	Role 4
Role 1	.	1.346	1.115	0.962
Role 2	1.038	.	1.654	1.192
Role 3	1.077	1.308	.	1.385
Role 4	1.462	1.269	1.346	.
Medium-inequality Baseline				
From:	Role 1	Role 2	Role 3	Role 4
Role 1	.	1.680	1.600	1.600
Role 2	1.120	.	1.480	1.280
Role 3	1.560	1.960	.	1.800
Role 4	1.360	1.640	1.680	.
High-inequality Baseline				
From:	Role 1	Role 2	Role 3	Role 4
Role 1	.	1.654	1.346	1.423
Role 2	1.577	.	1.692	1.615
Role 3	1.654	1.000	.	1.192
Role 4	1.423	1.269	1.269	.
Between-group inequality				
From:	Role 1	Role 2	Role 3	Role 4
Role 1	.	1.786	1.714	1.786
Role 2	1.536	.	1.250	1.250
Role 3	1.429	1.929	.	1.571
Role 4	2.036	2.250	2.036	.

Notes: Change in reported closeness is measured by the difference in the IoS scores reported towards a group member between the pre-task and post-task elicitations.

Table D2: (a) Change in Group Cohesion across the four pay levels in Between-group-inequality treatment

Pay Level	Main measure mean	Alternative 1 mean	Alternative 2 mean	Alternative 3 mean
Level 1	1.75	1.61	1.29	1.69
Level 2	1.86	1.43	1.29	1.80
Level 3	1.71	1.07	0.86	1.82
Level 4	1.57	1.29	0.86	1.55

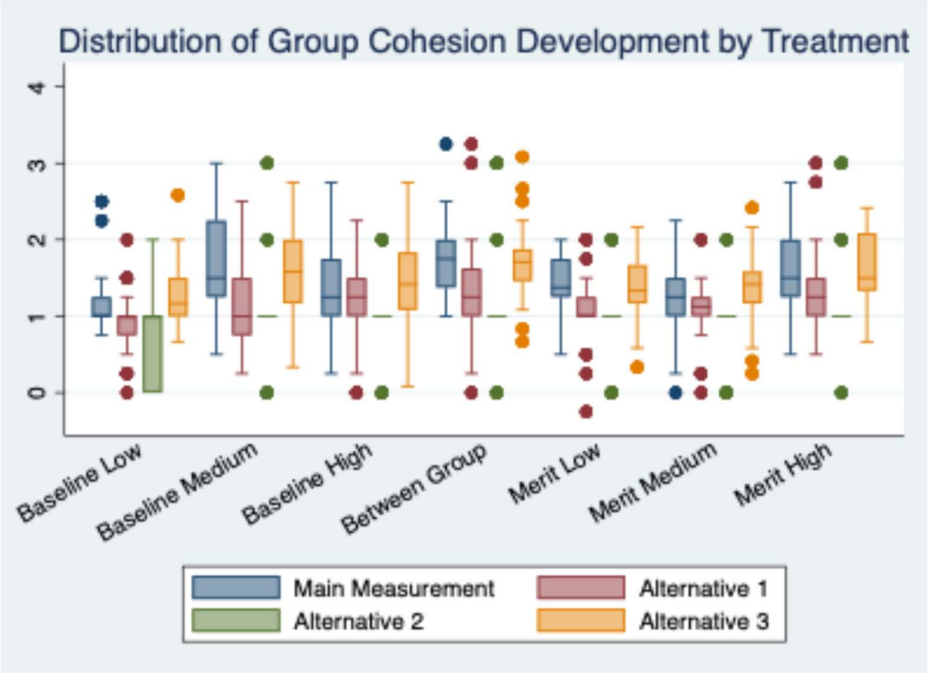
(b) Wilcoxon rank-sum tests

Pay level	Main measure	Alternative 1	Alternative 2	Alternative 3
comparison	p-value	p-value	p-value	p-value
Level 2 vs Level 1	0.476	0.467	1.000	0.370
Level 3 vs Level 1	0.795	0.471	0.297	0.438
Level 4 vs Level 1	0.694	0.462	0.173	1.000
Level 3 vs Level 2	0.515	0.895	0.297	0.748
Level 4 vs Level 2	0.217	1.000	0.173	0.522
Level 4 vs Level 3	0.435	0.948	0.936	0.607

*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

Appendix E: Dispersion in group cohesion development

Figure E1



Appendix E: Robustness Checks

Table E1: OLS Regressions on Change in Group Cohesion, by university

	Ningbo				Wuhan			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
	Model	Model	Model	Model	Model	Model	Model	Model
Treatment								
Baseline Low	ref.	ref.	ref.	ref.	ref.	ref.	ref.	ref.
	(.)	(.)	(.)	(.)	(.)	(.)	(.)	(.)
Baseline Medium	0.0424	0.0425	0.0780	0.0788	0.455*	0.455*	0.454*	0.453*
	(0.18)	(0.18)	(0.33)	(0.33)	(2.25)	(2.24)	(2.25)	(2.24)
Baseline High	-0.0588	-0.0580	-0.0825	-0.0812	0.167	0.167	0.172	0.172
	(-0.25)	(-0.25)	(-0.36)	(-0.35)	(0.85)	(0.85)	(0.88)	(0.88)
Between Group	0.366**	0.366**	0.349**	0.349**	0.509**	0.509**	0.512**	0.512**
	(3.19)	(3.20)	(3.01)	(3.05)	(2.89)	(2.89)	(2.89)	(2.88)
Merit Low	0.299*	0.299*	0.310*	0.310*	0.0401	0.0400	0.0387	0.0385
	(2.25)	(2.24)	(2.34)	(2.33)	(0.25)	(0.25)	(0.24)	(0.24)
Merit Medium	0.140	0.141	0.212	0.214	-0.0155	-0.0159	-0.0185	-0.0191
	(0.56)	(0.57)	(0.94)	(0.95)	(-0.10)	(-0.10)	(-0.12)	(-0.12)
Merit High	0.122	0.123	0.122	0.124	0.382*	0.381*	0.386*	0.385*
	(1.29)	(1.30)	(1.21)	(1.23)	(2.05)	(2.05)	(2.05)	(2.04)
Controls:								
Age	-0.00366	-0.00373	-0.00317	-0.00330	0.0207	0.0208	0.0212	0.0213
	(-0.24)	(-0.24)	(-0.21)	(-0.22)	(1.52)	(1.52)	(1.54)	(1.54)
Gender	0.0831	0.0815	-3.9×10^{-5}	-0.00441	-0.0178	-0.0166	0.00452	0.00634
	(1.22)	(1.18)	(-0.01)	(-0.56)	(-0.33)	(-0.30)	(0.68)	(0.67)
Slider Income		-5.3×10^{-5}		-1.1×10^{-4}		0.0000450		0.0000556
		(-0.32)		(-0.64)		(0.22)		(0.27)
Female proportion			0.369	0.374			-0.0836	-0.0852
			(1.21)	(1.22)			(-0.41)	(-0.42)
Constant	1.109**	1.118**	0.884*	0.900*	0.885**	0.879**	0.918**	0.910**
	(3.36)	(3.26)	(2.40)	(2.41)	(2.79)	(2.75)	(2.74)	(2.70)
N	202	202	202	202	532	532	532	532

Notes: The dependent variable is the change in group cohesion. Standard errors are clustered at the group level. t statistics in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table E2: Relationship between change in group cohesion and game outcomes, by university

	Ningbo		Wuhan	
	Game Outcome		Game Outcome	
	Public Goods	Weak Link	Public Goods	Weak Link
The change in group cohesion	1.161	-10.15	2.164	6.489
	(0.28)	(-1.11)	(1.69)	(1.39)
Control:				
Age	0.281	2.101	0.292	1.287
	(0.48)	(1.73)	(1.03)	(1.75)
Gender	0.476	1.835	-1.398	-5.399
	(0.18)	(0.37)	(-1.27)	(-1.56)
Constant	75.19***	95.94**	81.35***	102.5***
	(5.89)	(3.27)	(13.00)	(5.79)
Observations	190	202	532	532

Notes: OLS regressions are presented. The game outcomes are contributions in the public goods game and earnings in the weak-link game. t statistics in parentheses. Standard errors are clustered at the group level. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table E3: Treatment effects and mediating effects of group cohesion on game outcomes, by university

	Ningbo				Wuhan			
	Game Outcome		Game Outcome		Game Outcome		Game Outcome	
	Public Goods	Weak Link	Public Goods	Weak Link	Public Goods	Weak Link	Public Goods	Weak Link
Treatment:								
Baseline Low	ref.	ref.	ref.	ref.	ref.	ref.	ref.	ref.
	(.)	(.)	(.)	(.)	(.)	(.)	(.)	(.)
Baseline Medium	-2.458	-16.02	-2.975	-15.67	2.911	3.187	2.227	0.861
	(-0.45)	(-1.21)	(-0.52)	(-1.24)	(0.99)	(0.36)	(0.75)	(0.10)
Baseline High	-4.830	-20.77	-4.750	-21.27	0.408	0.489	0.157	-0.363
	(-0.96)	(-1.26)	(-0.94)	(-1.25)	(0.15)	(0.05)	(0.06)	(-0.04)
Between Group	-3.375	-25.37*	-3.901	-22.26	4.453	20.19*	3.688	17.59
	(-0.87)	(-2.02)	(-0.96)	(-1.76)	(1.60)	(2.05)	(1.27)	(1.68)
Merit Low	1.910	-15.18	1.483	-12.64	0.967	5.325	0.906	5.120
	(0.35)	(-0.98)	(0.27)	(-0.83)	(0.33)	(0.56)	(0.31)	(0.54)
Merit Medium	3.480	-27.47*	3.280	-26.28*	0.0502	5.048	0.0735	5.127
	(0.68)	(-2.15)	(0.63)	(-2.01)	(0.02)	(0.57)	(0.03)	(0.58)
Merit High	4.898	-26.40	4.734	-25.36	1.835	2.199	1.261	0.248
	(1.06)	(-1.81)	(1.01)	(-1.78)	(0.65)	(0.24)	(0.45)	(0.03)
The change in group cohesion			1.484	-8.485			1.504	5.111
			(0.37)	(-0.97)			(1.12)	(1.02)
Control:								
Age	0.542	1.633	0.566	1.602	0.345	1.567*	0.314	1.461*
	(1.03)	(1.70)	(1.09)	(1.68)	(1.22)	(2.08)	(1.09)	(1.99)
Gender	1.483	0.0541	1.364	0.759	-1.469	-5.459	-1.443	-5.368
	(0.67)	(0.01)	(0.61)	(0.15)	(-1.34)	(-1.63)	(-1.31)	(-1.60)
Constant	70.26***	113.2***	68.22***	122.6***	82.06***	101.2***	80.72***	96.65***
	(6.49)	(4.60)	(6.08)	(4.67)	(12.36)	(5.86)	(12.21)	(5.37)
Observations	190	202	190	202	532	532	532	532

Notes: OLS regressions are presented. The game outcomes are contributions in the public goods game and earnings in the weak-link game. t statistics in parentheses. Standard errors are clustered at the group level. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table E4: Percentage of subjects reporting that post-hoc redistribution was discussed

	Post-hoc redistribution discussed	
Treatment:	Yes	No
Baseline Low	10.6	89.4
Baseline Medium	7.0	93.0
Baseline High	6.7	93.3
Merit Low	3.8	96.2
Merit Medium	3.8	96.2
Merit High	8.3	91.7
Between Group	3.6	96.4
Total	6.3	93.7

Table E5: OLS Regressions on Change in Group Cohesion, excluding subjects who reported discussing post-hoc redistribution

	(1)	(2)	(3)	(4)
	Model	Model	Model	Model
Treatment				
Baseline Low	ref.	ref.	ref.	ref.
	(.)	(.)	(.)	(.)
Baseline Medium	0.391*	0.391*	0.391*	0.391*
	(2.33)	(2.33)	(2.36)	(2.36)
Baseline High	0.130	0.131	0.129	0.130
	(0.85)	(0.85)	(0.84)	(0.85)
Between Group	0.520***	0.520***	0.520***	0.520***
	(3.98)	(3.98)	(3.93)	(3.92)
Merit Low	0.165	0.165	0.165	0.165
	(1.40)	(1.40)	(1.40)	(1.40)
Merit Medium	0.0790	0.0802	0.0794	0.0809
	(0.64)	(0.65)	(0.65)	(0.66)
Merit High	0.351*	0.352*	0.350*	0.352*
	(2.54)	(2.54)	(2.53)	(2.53)
Controls:				
Age	0.0197	0.0195	0.0197	0.0195
	(1.53)	(1.52)	(1.53)	(1.51)
Gender	0.0155	0.0126	0.0140	0.0102
	(0.30)	(0.24)	(1.01)	(0.64)
Slider Income		-0.000101		-0.000102
		(-0.56)		(-0.55)
Female proportion			0.00598	0.00912
			(0.03)	(0.05)
Constant	0.780**	0.798**	0.777*	0.794*
	(2.71)	(2.74)	(2.52)	(2.56)

N	688	688	688	688

Notes: The dependent variable is the change in group cohesion. Standard errors are clustered at the group level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table E6: Treatment effects and mediating effects of group cohesion on game outcomes, excluding subjects who reported discussing post-hoc redistribution

	Game Outcome		Game Outcome		Game Outcome	
	Public Goods	Weak Link	Public Goods	Weak Link	Public Goods	Weak Link
The change in group cohesion	3.268*	4.080			3.076*	3.393
	(2.59)	(1.01)			(2.28)	(0.81)
Treatment:						
Baseline Low			ref.	ref.	ref.	ref.
			(.)	(.)	(.)	(.)
Baseline Medium			2.432	-3.540	0.878	-4.865
			(0.88)	(-0.44)	(0.32)	(-0.60)
Baseline High			-2.047	-9.826	-2.450	-10.27
			(-0.80)	(-1.13)	(-0.99)	(-1.18)
Between Group			1.720	5.388	0.123	3.622
			(0.64)	(0.62)	(0.05)	(0.40)
Merit Low			0.936	-2.126	0.434	-2.685
			(0.34)	(-0.25)	(0.16)	(-0.31)
Merit Medium			0.833	-5.371	0.593	-5.639
			(0.31)	(-0.67)	(0.23)	(-0.70)
Merit High			1.671	-8.083	0.590	-9.273
			(0.66)	(-0.96)	(0.23)	(-1.09)
Control:						
Age	0.504	1.459	0.565*	1.718*	0.518	1.651*
	(1.84)	(1.91)	(2.10)	(2.20)	(1.90)	(2.16)
Gender	-0.667	-3.418	-0.456	-3.222	-0.487	-3.275
	(-0.57)	(-1.13)	(-0.39)	(-1.09)	(-0.42)	(-1.10)
Constant	72.82***	99.52***	75.35***	103.0***	72.66***	100.4***
	(12.41)	(5.60)	(12.34)	(5.88)	(12.16)	(5.52)
Observations	676	688	676	688	676	688

Notes: OLS regressions are presented. The game outcomes are contributions in the public goods game and earnings in the weak-link game. t statistics in parentheses. Standard errors are clustered at the group level. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table E7: Significance of treatment differences for change in group cohesion – Wilcoxon rank-sum tests (subsamples)

	Subsample of		
	Ningbo	Wuhan	Negative Answer to Redistribution
Hypothesis 1			
Low-inequality Baseline vs Medium-inequality Baseline	0.530	0.011*	0.018*
Low-inequality Baseline vs High-inequality Baseline	0.669	0.229	0.184
Medium-inequality Baseline vs High-inequality Baseline	0.885	0.286	0.334
Low-inequality Merit-based vs Medium-inequality Merit-based	0.513	0.698	0.533
Low-inequality Merit-based vs High-inequality Merit-based	0.185	0.071	0.401
Medium-inequality Merit-based vs High-inequality Merit-based	0.631	0.026*	0.174
Hypothesis 2			
Low-inequality Baseline vs Between-group-inequality	0.010**	0.004**	0.000***
Hypothesis 3			
Low-inequality Baseline vs Low-inequality Merit-based	0.041*	0.315	0.035*
Medium-inequality Baseline vs Medium-inequality Merit-based	0.470	0.074	0.299
High-inequality Baseline vs High-inequality Merit-based	0.809	0.263	0.277

Notes: p-values are presented. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. *For the* Negative Answer to Redistribution *subsample, we run cluster-adjusted tests (using the STATA command Somersd), using individual-level observations for change in group cohesion and clustering at the group level. This accounts for the problem that* Negative Answer to Redistribution *is an individual-level variable which can vary within groups.*